NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

CE-323 – Quantity & Cost Estimation

COMPLEX ENGINEERING PROBLEM

DEPARTMENT OF CIVIL ENGINEERING

GROUP MEMBERS:

Muhammad Aqib Hameed	**CE-22011**
Hafiz Muhammad Yaseen	**CE-22014**
Muhammad Zaryab Ali	**CE-22016**
Syed Muhammad Ali Mehdi	**CE-22043**

CONSTRUCTION PROJECT COST SUMMARY

Project: Construction of a 500 sq.yd G+1 Residential Building (Civil Works Only)

Location: Plot # 35, Road # 9, Bahria Town, Karachi

Bill of Quantities: Main Summary of Cost

S. No.	Description	Amount (Rs)
1	Civil Works	13,466,012
	G. TOTAL	**13,466,012**

Total Cost in Words: Thirteen Million,four hundred sixty six Thousand, and Twelve Only/-

Contractor Signatur ____________________

Consultant Signature: ____________________

CE-323 Quantity and Cost Estimation

Department of Civil Engineering

NED University of Engineering and Technology

COMPLEX ENGINEERING PROBLEM

Batch: 2022

Group Member Names:

Muhammad Aqib Hameed	CE-22011
Hafiz Muhammad Yaseen	CE-22014
Muhammad Zaryab Ali	CE-22016
Syed Muhammad Ali Mehdi	CE-22043

Approved by

……………………………………………………………………………………………

Dr. Rana Rabnawaz Ahmed	Dr. Farrukh Arif
Assistant Professor	Assistant Professor
Course Teacher	Course Teacher

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Certificate

This is to certify that the following students of Batch 2022 have successfully completed the Complex Engineering Problem (CEP) in Quantity and Cost Estimation.

Batch: 2022

Group Member Names:

Muhammad Aqib Hameed	CE-22011
Hafiz Muhammad Yaseen	CE-22014
Muhammad Zaryab Ali	CE-22016
Syed Muhammad Ali Mehdi	CE-22043

COURSE TEACHER

____

Dr. Rana Rabnawaz
Ahmed
Assistant Professor
Department of Civil Engineering

Author’s Declaration

We confirm that this CEP has been written entirely byus. This is the final version that was reviewed and approved by our advisor(s), including all required changes. We also allow NED University of Engineering and Technology to print or share copies of this project in digital or physical form.

Signature and Date	Signature and Date	Signature and Date	Signature and Date
....	....	....	....
Muhammad Aqib Hameed	Hafiz Muhammad Yaseen	Muhammad Zaryab Ali	Syed Muhammad Ali Mehdi
CE-22011	CE-22014	CE-22016	CE-22043

Statement of Contributions

All group members had supplied their efforts equally in the Literature review and other additional miscellaneous tasks.

- Muhammad Aqib Hameed actively contributed to Bill of Quantities, Specifications, Slab Concreting, Plaster quantites, Stair case Reinforcement and report writing.
- Muhammad Zaryab Ali activelyparticipated in complete sub-structure, Super-structure (Slab Reinforcement), Columns, Stair case concrete.
- Hafiz Muhammad Yaseen contribute in reinforcement of beams (First and roof) And plinth beam, wooden work complete, OHWT and UGWT (Reinforcement and concreting)
- Syed Muhammad Ali Mehdi actively participated in Block masonary, Marble and flooring and Rate analysis

BILL OF QUANTITES					
S.NO.	Descritption	Quantity	Unit	Rate (RS)	Amount (Rs)
1	**Site Preparation**	4500	sft	12.711	57200
	Activity: Site clearing, grubbing, and removal of vegetation; establishment of survey control points and site layout. Details: Includes all operations necessary to prepare the site as per the Engineer-in-Charge's directives. This encompasses clearing of bushes, setting up burjies, and conducting initial surveys.				
2	**Termite proofing**	2925	Liters	57.8	169065
	Termite protection treatment applied to foundation, trenches, columns, floors, and wooden structures using Bioflex Pro solution, as per manufacturer's instructions and Engineer's supervision. Payment for ground floor plinth area treatment only.				
2	**Earthworks**				
2.1	**Foundation, Boundary wall & UGWT Excavation**	7464.28	cft	8.395	62662.63
	Description: Excavation to specified depths from Natural Ground Level (NGL) as per drawings for foundations and UGWT. Specifications: All soil types; backfilling with excavated material in 9-inch compacted layers; surface dressing; disposal of excess soil. Repair any damage to utilities.				
2.2	**Imported Earth Fill**	3663.54	cft	23.5	25018.638
	Description: Supply and placement of approved fill material. Specifications: Selected soil from off-site; placement in foundations, trenches, plinths; clod breaking, leveling, watering, and compaction in 9-inch layers. Architect/Engineer-in-Charge approval required.				

3	**Sub-Structure Construction**				
3.1	**Lean Concrete**	1115.29	cft	328.04	365861
	Description: Placement of lean concrete mix. Specifications: 15 mpa (cement:sand:aggregate); 4-inch thickness in footings, walls and UGWT areas as per drawings.				
3.2	**Sub-Structure Reinforcement**	4.96	tons	258800	1283648
	Description: Reinforcement for RCC foundation work. Specifications: Straightening, cutting, bending, placing, and binding steel bars to structural details; 60 ksi yield strength.				
3.3	**Sub-Structure Concreting**	1465.57	cft	397	582980
	Description: Reinforced concrete work for sub-structure. Specifications: 1:2:4 concrete ratio; includes formwork, binding wire, and two coats of bitumen. Architect consultation required				
4	**Super-Structure Construction**				
4.1	**Super-Structure Reinforcement**	9.458	tons	258800	2447730.4
	Description: Reinforcement for RCC superstructure elements. Specifications: Floor beams, columns, roof beams, slabs (ground and first floors), and overhead water tank; straightening, cutting, bending, placing, and binding bars; 60 ksi yield strength.				
4.2	**Super-Structure Concreting**	6057.864	cft	554	3357746
	Description: Reinforced concrete work. Specifications: Plinth beams, columns, roof beams,floor beam, OHWT, slabs, and staircases (ground and first floors); 1:2:4 concrete for slabs/beams; 1:1.5:3 concrete for columns; includes shuttering, binding wire, and spacers. Execution as per drawings and Engineer-in-Charge.				

5	**Masonry**	7840	blocks	114	893760
	Description: Blockwork for walls. Specifications: 6" x 9" x 12" blocks; 2" DPC on boundary and exterior walls; 1:4 mortar mix; stretcher bond; 1/2" joints; ground and first floors.				
6	**Finishes**	26908.9	cft	49.29	1326475
	Description: Plaster application. Specifications: 1/2" plaster thickness; 1:3 plaster ratio; boundary walls, exterior facades, interior of ground and first floors; Consultant Architect approval required.				
7	**Flooring**				
7.1	**Marble Flooring**	4943.69	sft	412.5	2039272.125
	Description: Marble flooring installation. Specifications: Ground floor, first floor, and staircase (excluding bathrooms/powder room); 2" PCC underlayer; grouting, scrubbing, polishing; skirting.				
7.2	**Tile Flooring**	1292.5	sft	446	576455
	Description: Tile flooring installation. Specifications: Bathrooms and powder rooms (ground and first floors); white cement grout; Supervisor approval required.				
8	**Doors, Windows and Ventilators**	4383.4	rft	63.45	278139
	Doors				
	D1(4'-6" x 7'-0")				
	D2(3'-6" x 7'-0")				
	D3(3'-0" x 7'-0")				
	D4(2'-6" x 7'-0")				
	Sliding door(9'-6" x 7'-0")				
	Windows				
	W1(10'-0" x 4'-0")				
	W2(8'-0" x 4'-0")				
	W3(5'-0" x 4'-0")				
	W4(9'-6" x 4'-0")				
	W5(4'-0" x 4'-0")				
	Ventilators				
	V1(2'-0" x 1'-6")				
	Main Gate(Steel/2panel)				
	Main Gate(Steel/sigle panel)				
	TOTAL				**13466012.79**

MEASUREMENT SHEET (MS)

S NO	Description Of Work	Number	Length (ft)	Breadth (ft)	Height (ft)	Quantity (cft)	Total Quantity (cft)
	Sub-Structure (Excavation,Lean,Situ and Concreting)						
1	**Excavation**						
1.1	**Excavation for footing**						
	F-1	11	5.00	4.50	5.00	112.50	1237.50
	F-2	13	6.00	5.50	5.00	165.00	2145.00
	F-3	1	3.00	6.00	5.00	90.00	90.00
	F-4	22	3.50	4.00	5.00	70.00	1540.00
	F-5	3	3.50	3.50	5.00	61.25	183.75
	F-6	5	5.00	5.50	5.00	137.50	687.50
	F-7	0	3.50	3.50	5.00	61.25	0.00
	CF-1	1	6.50	3.50	5.00	113.75	113.75
1.2	**Excavation for Boundary Wall**						
	Longer Side 01 (Left)	1	51.25	1.50	2.00	153.75	153.75
	Longer Side 02 (Right)	1	49.75	1.50	2.00	149.25	149.25
	Shorter Side 01 (Back)	1	28.00	1.50	2.00	84.00	84.00
	Shorter Side 02 (Front)	1	19.67	1.50	2.00	59.01	59.01
1.3	**Excavation for External Wall**						
	Grid A (with offset wall and grid D,G)	1	35.79	1.50	2.00	107.38	107.38
	Grid P	1	41.92	1.50	2.00	125.75	125.75
	Grid 1	1	14.00	1.50	2.00	42.00	42.00
	Grid 14 (also grid 11 and 12)	1	16.25	1.50	2.00	48.75	48.75
1.4	**Excavation for UGWT**						
	UGWT	1	9.33	9.33	8.00	696.89	696.89
1.5	**Total Excavation**						**7464.28**
2	**1:4:8 Lean Concrete**						
2.1	**Lean for footing**						
	F-1	11	5.00	4.50	0.50	11.25	123.75
	F-2	13	6.00	5.50	0.50	16.50	214.50
	F-3	1	3.00	6.00	0.50	9.00	9.00
	F-4	22	3.50	4.00	0.50	7.00	154.00
	F-5	3	3.50	3.50	0.50	6.13	18.38
	F-6	5	5.00	5.50	0.50	13.75	68.75
	F-7	0	3.50	3.50	0.50	6.13	0.00
	CF-1	1	6.50	3.50	0.50	11.38	11.38
2.2	**Lean for UGWT**						
	UGWT Lean	1	9.33	7.33	0.33	22.81	22.81
2.3	**Lean for boundary wall**						
	Longer Side 01 (Left)	1	78.25	1.50	0.50	58.69	58.69
	Longer Side 02 (Right)	1	78.75	1.50	0.50	59.06	59.06
	Shorter Side 01 (Back)	1	45.00	1.50	0.50	33.75	33.75
	Shorter Side 02 (Front)	1	32.00	1.50	0.50	24.00	24.00
2.4	**Lean for long wall**						
	Grid A (with offset wall)	1	50.58	1.50	0.50	37.94	37.94
	Grid B-1	1	4.83	1.50	0.50	3.63	3.63
	Grid C	1	5.83	1.50	0.50	4.38	4.38
	Grid D	1	14.00	1.50	0.50	10.50	10.50
	Grid F	1	41.00	1.50	0.50	30.75	30.75
	Grid G	1	7.17	1.50	0.50	5.38	5.38
	Grid H	1	17.00	1.50	0.50	12.75	12.75
	Grid J	1	14.50	1.50	0.50	10.87	10.87
	Grid K (offset wall)	1	7.00	1.50	0.50	5.25	5.25
	Grid M	1	13.50	1.50	0.50	10.13	10.13
	Grid P	1	60.67	1.50	0.50	45.50	45.50
2.5	**Lean for short wall**						

	Grid 1	1	31.50	1.50	0.50	23.63	23.63
	Grid 2	1	13.00	1.50	0.50	9.75	9.75
	Grid 3-A (Offset wall)	1	5.00	1.50	0.50	3.75	3.75
	Grid 3	1	18.25	1.50	0.50	13.69	13.69
	Grid 4	1	11.50	1.50	0.50	8.63	8.63
	Grid 6-A	1	4.58	1.50	0.50	3.44	3.44
	Grid 6	1	13.50	1.50	0.50	10.13	10.13
	Grid 8	1	24.00	1.50	0.50	18.00	18.00
	Grid 9	1	9.50	1.50	0.50	7.13	7.13
	Grid 9 (Angled wall)	1	9.30	1.50	0.50	6.98	6.98
	Grid 10	1	14.00	1.50	0.50	10.50	10.50
	Grid 11	1	7.33	1.50	0.50	5.50	5.50
	Grid 12	1	8.17	1.50	0.50	6.13	6.13
	Grid 14	1	17.25	1.50	0.50	12.94	12.94
2.5	**Total Lean concrete**						**1115.29**
3	**1:3:6 Situ below Plinth**						
3.1	**Situ in Boundary wall**						
	Longer Side 01 (Left)	1	78.25	1.50	1.50	176.06	176.06
	Longer Side 02 (Right)	1	78.75	1.50	1.50	177.19	177.19
	Shorter Side 01 (Back)	1	45.00	1.50	1.50	101.25	101.25
	Shorter Side 02 (Front)	1	32.00	1.50	1.50	72.00	72.00
3.2	**Situ in External wall**						
	Grid A (with offset wall and grid D,G)	1	57.75	1.50	1.50	129.94	129.94
	Grid P	1	60.67	1.50	1.50	136.50	136.50
	Grid 1	1	31.50	1.50	1.50	70.88	70.88
	Grid 14 (also grid 11 and 12)	1	30.25	1.50	1.50	68.06	68.06
3.3	**Total Situ**						**931.88**
4	**1:2:4 Concrete**						
4.1	**1:2:4 Concrete for footing**						
	F-1	11	4.00	3.50	1.50	21.00	231.00
	F-2	13	5.00	4.50	1.50	33.75	438.75
	F-3	1	2.00	5.00	1.50	15.00	15.00
	F-4	22	2.50	3.00	1.50	11.25	247.50
	F-5	3	2.50	2.50	1.50	9.38	28.13
	F-6	5	4.00	4.50	1.50	27.00	135.00
	F-7	0	2.50	2.50	1.50	9.38	0.00
	CF-1	1	5.50	2.50	1.50	20.63	20.63
	Total 1:2:4 Concrete for footing						**1116.00**
4.2	**1:2:4 Concrete for Column below Plinth**						
	C-1	18	1.50	0.50	3.00	2.25	40.50
	C-2	6	1.75	0.50	3.00	2.63	15.75
	C-3	7	1.17	0.50	3.00	1.75	12.25
	C-4	2	2.00	0.50	3.00	3.00	6.00
	C-5	1	1.75	0.67	3.00	3.50	3.50
	C-6	1	1.75	0.67	3.00	3.50	3.50
	C-7	22	1.00	0.50	3.00	1.50	33.00
	C-8	3	1.00	1.00	3.00	3.00	9.00
	Total 1:2:4 Concrete for Column below Plinth						**123.50**
4.3	**1:2:4 Concrete for UGWT**						
	Bottom Slab	1	9.33	7.33	1.00	68.44	68.44
	Top Slab	1	9.33	7.33	0.67	45.63	45.63
	Longer Wall	2	8.00	0.67	6.00	32.00	64.00
	Shorter Wall	2	6.00	0.67	6.00	24.00	48.00
	Total 1:2:4 Concrete for UGWT						**226.07**
4.4	**Total 1:2:4 Concrete in Sub-Structure**						**1465.57**

4.5	**Backfill of Soil**						
	EXC-Lean Concrete-Situ-1:2:4 Concrete-UGWT VOL						**3663.54**

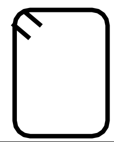
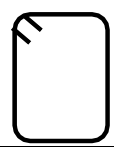
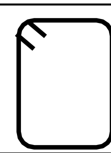
MEASUREMENT SHEET (MS)

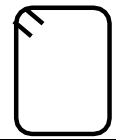
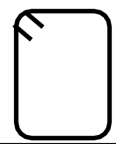
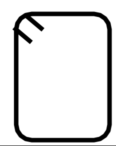
S NO	Description Of Work	Number	Length (ft)	Breadth (ft)	Height (ft)	Quantity (cft)	Total Quantity (cft)
	Super-Structure (Concreting)						
1	1:2:4 Concrete for Beams						
1.1	**1:2:4 Concrete for Plinth Beams**						
	Boundar wall PB-01	1	266.33	0.50	2.00	266.33	266.33
	Boundar wall PB-04	1	11.17	0.67	2.00	14.89	14.89
	Grid A (with offset beam)	1	59.25	0.50	2.00	59.25	59.25
	Grid B-1	1	6.00	0.50	2.00	6.00	6.00
	Grid C	1	7.00	0.50	2.00	7.00	7.00
	Grid D	1	15.50	0.50	2.00	15.50	15.50
	Grid F	1	46.25	0.50	2.00	46.25	46.25
	Grid G	1	8.17	0.50	2.00	8.17	8.17
	Grid H	1	20.50	0.50	2.00	20.50	20.50
	Grid J	1	17.00	0.50	2.00	17.00	17.00
	Grid K (offset beam)	1	7.00	0.50	2.00	7.00	7.00
	Grid M	1	15.00	0.50	2.00	15.00	15.00
	Grid P	1	68.92	0.50	2.00	68.92	68.92
	Grid 1	1	33.50	0.50	2.00	33.50	33.50
	Grid 2	1	13.00	0.50	2.00	13.00	13.00
	Grid 3-A (Offset beam)	1	6.00	0.50	2.00	6.00	6.00
	Grid 3	1	20.50	0.50	2.00	20.50	20.50
	Grid 4	1	12.50	0.50	2.00	12.50	12.50
	Grid 6-A	1	5.58	0.50	2.00	5.58	5.58
	Grid 6	1	15.50	0.50	2.00	15.50	15.50
	Grid 8	1	28.00	0.50	2.00	28.00	28.00
	Grid 9	1	11.50	0.50	2.00	11.50	11.50
	Grid 9 (Angled beam)	1	9.30	0.50	2.00	9.30	9.30
	Grid 10	1	15.50	0.50	2.00	15.50	15.50
	Grid 11	1	8.83	0.50	2.00	8.83	8.83
	Grid 12	1	9.17	0.50	2.00	9.17	9.17
	Grid 14	1	19.50	0.50	2.00	19.50	19.50
	Total 1:2:4 Concrete for Plinth Beam						**760.19**
1.1	**1:2:4 Concrete for floor Beams**						
	Grid A-1 (offset beam)	1	12.17	0.50	2.50	15.21	15.21
	Grid A (with offset beam)	1	59.25	0.50	2.50	74.06	74.06
	Grid B-1	1	6.00	0.50	2.50	7.50	7.50
	Grid C	1	7.00	0.50	2.50	8.75	8.75
	Grid D	1	15.50	0.50	2.50	19.38	19.38
	Grid F	1	46.25	0.50	2.50	57.81	57.81
	Grid G	1	8.17	0.50	2.50	10.21	10.21
	Grid H	1	20.50	0.50	2.50	25.63	25.63
	Grid J	1	17.00	0.50	2.50	21.25	21.25
	Grid K (offset beam)	1	7.00	0.50	2.50	8.75	8.75
	Grid M	1	15.00	0.50	2.50	18.75	18.75
	Grid P	1	68.92	0.50	2.50	86.15	86.15
	Grid 1	1	33.50	0.50	2.50	41.88	41.88
	Grid 2	1	13.00	0.50	2.50	16.25	16.25
	Grid 3-A (Offset beam)	1	6.00	0.50	2.50	7.50	7.50
	Grid 3	1	20.50	0.50	2.50	25.63	25.63
	Grid 4	1	12.50	0.50	2.50	15.63	15.63
	Grid 6-A	1	5.58	0.50	2.50	6.98	6.98
	Grid 6	1	15.50	0.50	2.50	19.38	19.38
	Grid 8	1	28.00	0.50	2.50	35.00	35.00
	Grid 9	1	11.50	0.50	2.50	14.38	14.38

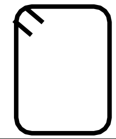
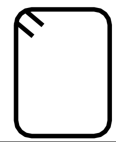
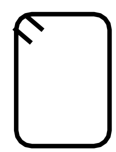
	Grid 9 (Angled beam)	1	9.30	0.50	2.50	11.63	11.63
	Grid 10	1	15.50	0.50	2.50	19.38	19.38
	Grid 11	1	15.42	0.50	2.50	19.27	19.27
	Grid 12	1	9.17	0.50	2.50	11.46	11.46
	Grid 14	1	19.50	0.50	2.50	24.38	24.38
	Grid 14-A (Projecting beam	1	18.33	0.50	2.50	22.91	22.91
	Total 1:2:4 Concrete for floor Beams						**645.06**
1.3	**1:2:4 Concrete for Roof Beams**						
	Boundar wall PB-01	1	266.33	0.50	2.50	332.92	332.92
	Boundar wall PB-04	1	11.17	0.67	2.50	18.61	18.61
	Grid A (with offset beam)	1	59.25	0.50	2.50	74.06	74.06
	Grid B-1	1	6.00	0.50	2.50	7.50	7.50
	Grid C	1	7.00	0.50	2.50	8.75	8.75
	Grid D	1	15.50	0.50	2.50	19.38	19.38
	Grid F	1	46.25	0.50	2.50	57.81	57.81
	Grid G	1	8.17	0.50	2.50	10.21	10.21
	Grid H	1	20.50	0.50	2.50	25.63	25.63
	Grid J	1	17.00	0.50	2.50	21.25	21.25
	Grid K (offset beam)	1	7.00	0.50	2.50	8.75	8.75
	Grid M	1	15.00	0.50	2.50	18.75	18.75
	Grid P	1	68.92	0.50	2.50	86.15	86.15
	Grid 1	1	33.50	0.50	2.50	41.88	41.88
	Grid 2	1	13.00	0.50	2.50	16.25	16.25
	Grid 3-A (Offset beam)	1	6.00	0.50	2.50	7.50	7.50
	Grid 3	1	20.50	0.50	2.50	25.63	25.63
	Grid 4	1	12.50	0.50	2.50	15.63	15.63
	Grid 6-A	1	5.58	0.50	2.50	6.98	6.98
	Grid 6	1	15.50	0.50	2.50	19.38	19.38
	Grid 8	1	28.00	0.50	2.50	35.00	35.00
	Grid 9	1	11.50	0.50	2.50	14.38	14.38
	Grid 9 (Angled beam)	1	9.30	0.50	2.50	11.63	11.63
	Grid 10	1	15.50	0.50	2.50	19.38	19.38
	Grid 11	1	8.83	0.50	2.50	11.04	11.04
	Grid 12	1	9.17	0.50	2.50	11.46	11.46
	Grid 14	1	19.50	0.50	2.50	24.38	24.38
	Total 1:2:4 Concrete for Roof Beam						**950.24**
1.4	**Total 1:2:4 Concrete for Beams**						**2355.49**
1	**1:2:4 Concrete for Columns above Plinth Beam**						
1.1	**1:2:4 Concrete for Column from Plinth level till first floor**						
	C-1	18	8.00	0.50	3.00	12.00	216.00
	C-2	6	8.00	0.50	3.00	12.00	72.00
	C-3	7	8.00	0.50	3.00	12.00	84.00
	C-4	2	8.00	0.50	3.00	12.00	24.00
	C-5	1	8.00	0.67	3.00	16.00	16.00
	C-6	1	8.00	0.67	3.00	16.00	16.00
	C-7	22	8.00	0.50	3.00	12.00	264.00
	C-8	3	8.00	1.00	3.00	24.00	72.00
	Total 1:2:4 Concrete for Column						**764.00**
1.2	**1:2:4 Concrete for Column from first floor till Roof**						
	C-1	18	8.00	0.50	3.00	12.00	216.00
	C-2	6	8.00	0.50	3.00	12.00	72.00
	C-3	7	8.00	0.50	3.00	12.00	84.00

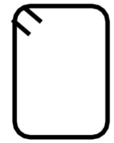
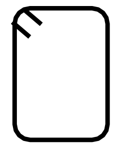
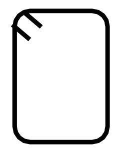
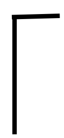
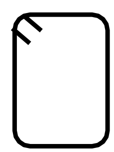
	C-4	2	8.00	0.50	3.00	12.00	24.00
	Total 1:2:4 Concrete for Column						**396.00**
	1:2:4 Concrete for Slabs						
1	**Roof Slab**						
	Total Area	1	68.92	36.00	0.50	1240.52	1240.52
	Deduction 01	1	9.6676	9.333	0.50	45.11	45.11
	Deduction 02	1	7.6676	4.167	0.50	15.98	15.98
	Deduction 03	1	7	5.5	0.50	19.25	19.25
	Deduction 04 (Stairs)	1	13	9.333	0.50	60.66	60.66
	Total Roof Slab (Total - All deductions)						1099.513
2	**First floor Slab**						
	Total Area	1	68.92	36.00	0.50	1240.52	1240.52
	Deduction 01	1	9.6676	9.333	0.50	45.11	45.11
	Deduction 02	1	7.6676	4.167	0.50	15.98	15.98
	Deduction 03	1	7	5.5	0.50	19.25	19.25
	Deduction 04 (Stairs)	1	13	9.333	0.50	60.66	60.66
	Total Roof Slab (Total + (∑deductions 01 to 03) - Deduction 04)						1260.19
1.4	**Concrete of ohwt**						
	Base Slab	1	7.333	7.333	0.667	35.86651696	35.86651696
	Long Wall	2	7.333	0.667	5	24.455555	48.91111
	Short Wall	2	6	0.667	5	20.01	40.02
	Top Slab	1	7.333	7.333	0.667	35.86651696	35.86651696
	Base Beam	2	7.333	0.667	2.25	11.00499975	22.0099995
	Total						182.6741434
	Concrete for Staircase						
	Landing Slab	2	9.33	4	0.5	18.66	37.32
	Waist Slab (sqrt(13^2+5^2)) = 13.928	4	13.928	4.665	0.5	32.48706	129.94824
	Steps (Triangle Area * Length) Riser 6", Tread 1.5'	40	1.5	1.5	0.5	1.125	45
	Total Concrete in Stairs						

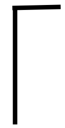
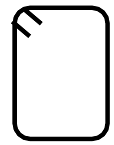
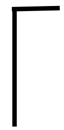
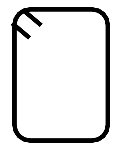
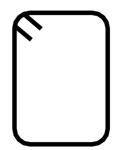
S NO	Description Of Work	Number	Bar Number	Diameter (in)	Shape	SL (ft)	HL (ft) (12Dia)	Total Length (ft)	Wt/ft	Weight (lbs)	Weight (tonnes)
	BAR BENDING SCHEDULE (BBS)										
1	**R/f in footing**										
1.1	**Footing F-1** (4x3.5)										
	Main Bar (#4@6"c/c)	94	4	0.5		3.25	1	4.25	0.67	264.92	0.12
	Distribution Bar (#4@6"c/c)	83	4	0.5		3.75	1	4.75	0.67	261.25	0.12
1.2	**Footing F-2** (5x4.5)										
	Main Bar (#4@6"c/c)	137	4	0.5		4.25	1	5.25	0.67	477.75	0.22
	Distribution Bar (#4@6"c/c)	124	4	0.5		4.75	1	5.75	0.67	473.42	0.21
1.3	**Footing F-3** (4x5)										
	Main Bar (#4@6"c/c)	11	4	0.5		3.75	1	4.75	0.67	33.25	0.02
	Distribution Bar (#4@6"c/c)	9	4	0.5		4.75	1	5.75	0.67	32.58	0.01
1.4	**Footing F-4** (2.5x3)										
	Main Bar (#4@6"c/c)	143	4	0.5		2.25	1	3.25	0.67	309.83	0.14
	Distribution Bar (#4@8"c/c)	96	4	0.5		2.75	1	3.75	0.67	240.63	0.11
1.5	**Footing F-5** (2.5x2.5)										
	Main Bar (#4@6"c/c)	17	4	0.5		2.25	1	3.25	0.67	35.75	0.02
	Distribution Bar (#4@6"c/c)	17	4	0.5		2.25	1	3.25	0.67	35.75	0.02
1.6	**Footing F-6** (4x4.5)										
	Main Bar (#4@6"c/c)	48	4	0.5		3.75	1	4.75	0.67	150.42	0.07
	Distribution Bar (#4@6"c/c)	43	4	0.5		4.25	1	5.25	0.67	148.75	0.07
1.7	**Footing F-7** (2.5x2.5)										
	Main Bar (#4@6"c/c)	0	4	0.5		2.25	1	3.25	0.67	0.00	0.00
	Distribution Bar (#4@6"c/c)	0	4	0.5		2.25	1	3.25	0.67	0.00	0.00
1.8	**Footing CF-1** (2.5x5.5)										
	Main Bar (#4@8"c/c)	9	4	0.5		2.25	1	3.25	0.67	19.23	0.01

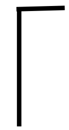
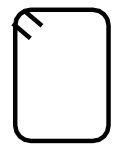
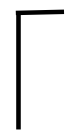
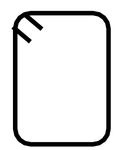
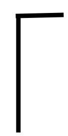
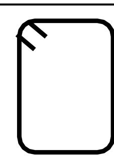
	Distribution Bar (#4@8"c/c)	4	4	0.5		5.25	1	6.25	0.67	18.23	0.01
	Total R/f in footings									**2501.75**	**1.14**
2	**R/f in Column from Foundation till Plinth**										
2.1	**Column C-1**										
	Dowel Bars 5'+48(0.5)" =7'	108	4	0.5		7	0.5	7.50	0.67	540.00	0.25
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	162	3	0.375		3	0.5	3.50	0.38	212.63	0.10
2.2	**Column C-2**										
	Dowel Bars 5'+48(0.5)" =7'	36	4	0.5		7	0.5	7.50	0.67	180.00	0.08
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	54	3	0.375		3.5	0.5	4.00	0.38	81.00	0.04
2.3	**Column C-3**										
	Dowel Bars 5'+48(0.5)" =7'	28	4	0.5		7	0.5	7.50	0.67	140.00	0.06
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	63	3	0.375		2.33	0.5	2.83	0.38	66.94	0.03
2.4	**Column C-4**										

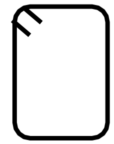
	Dowel Bars 5'+48(0.5)" =7'	16	4	0.5		7	0.5	7.50	0.67	80.00	0.04
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	18	3	0.375		4	0.5	4.50	0.38	30.38	0.01
2.5	**Column C-5**										
	Dowel Bars 5 +48(0.5)" =7'	6	4	0.5		7	0.5	7.50	0.67	30.00	0.01
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	9	3	0.375		3.83	0.5	4.33	0.38	14.63	0.01
2.6	**Column C-6**										
	Dowel Bars 5'+48(0.5)" =7'	8	4	0.5		7	0.5	7.50	0.67	40.00	0.02
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@6" (Span/Spacing)+1= 11 levels	11	3	0.375		3.83	0.5	4.33	0.38	17.88	0.01
2.7	**Column C-7**										
	Dowel Bars 5'+48(0.5)" =7'	88	4	0.5		7	0.5	7.50	0.67	440.00	0.20

	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	198	3	0.375		2	0.5	2.50	0.38	185.63	0.08
2.8	**Column C-8**										
	Dowel Bars 5'+48(0.5)" =7'	12	4	0.5		7	0.5	7.50	0.67	60.00	0.03
	Ties up to Plnth (3 ft column below PB and above footing +2 of PB =5ft@8" (Span/Spacing)+1= 9 levels	27	3	0.375		3	0.5	3.50	0.38	35.44	0.02
	Total R/f									**2154.50**	**0.98**
3	**R/f in Column from Plinth till first floor**										
3.1	**Column C-1**										
	Dowel Bars 10'6" +48(0.5)" =12.5'	108	4	0.5		12.5	0.5	13.00	0.67	936.00	0.42
	Ties up to first floor (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	288	3	0.375		3	0.5	3.50	0.38	378.00	0.17
3.2	**Column C-2**										
	Dowel Bars 10'6" +48(0.5)" =12.5'	36	4	0.5		12.5	0.5	13.00	0.67	312.00	0.14

	Ties up to first floor (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	96	3	0.375		3.5	0.5	4.00	0.38	144.00	0.07
3.3	**Column C-3**										
	Dowel Bars 10'6" +48(0.5)" =12.5'	28	4	0.5		12.5	0.5	13.00	0.67	242.67	0.11
	Ties up to first floor (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	112	3	0.375		2.33	0.5	2.83	0.38	119.00	0.05
2.4	**Column C-4**										
	Dowel Bars 10'6" +48(0.5)" =12.5'	16	4	0.5		12.5	0.5	13.00	0.67	138.67	0.06
	Ties up to first floor (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	32	3	0.375		4	0.5	4.50	0.38	54.00	0.02
2.5	**Column C-5**										
	Dowel Bars 10'6" +12(0.5)" =11'	6	4	0.5		11	0.5	11.50	0.67	46.00	0.02
	Ties up to first floor (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	16	3	0.375		3.83	0.5	4.33	0.38	25.98	0.01
2.6	**Column C-6**										

	Dowel Bars 10'6" +12(0.5)" =11'	8	4	0.5		11	0.5	11.50	0.67	61.33	0.03
	Ties up to first floor (10'6"-6") =10'@6 (Span/Spacing)+1 =21 levels	21	3	0.375		3.83	0.5	4.33	0.38	34.10	0.02
2.7	Column C-7										
	Dowel Bars 8 +12(0.5)" =8.5'	88	4	0.5		8.5	0.5	9.00	0.67	528.00	0.24
	Ties up to first floor (8') =8'@8 (Span/Spacing)+1 =13 levels	286	3	0.375		2	0.5	2.50	0.38	268.13	0.12
2.8	Column C-8										
	Dowel Bars 8 +12(0.5)" =8.5'	12	4	0.5		8.5	0.5	9.00	0.67	72.00	0.03
	Ties up to first floor (8') =8'@8 (Span/Spacing)+1 =13 levels	39	3	0.375		3	0.5	3.50	0.38	51.19	0.02
	Total R/f									**3411.06**	**1.55**
3	**R/f in Column from first floor till Roof**										
3.1	Column C-1										

	Dowel Bars 10'6" +12(0.5)" =11'	108	4	0.5		11	0.5	11.5	0.67	828.00	0.38
	Ties up to roof (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	288	3	0.375		3	0.5	3.5	0.38	378.00	0.17
3.2	**Column C-2**										
	Dowel Bars 10'6" +12(0.5)" =11'	36	4	0.5		11	0.5	11.5	0.67	276.00	0.13
	Ties up to roof (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	96	3	0.375		3.5	0.5	4	0.38	144.00	0.07
3.3	**Column C-3**										
	Dowel Bars 10'6" +12(0.5)" =11'	28	4	0.5		11	0.5	11.5	0.67	214.67	0.10
	Ties up to roof (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	112	3	0.375		2.33	0.5	2.83	0.38	118.86	0.05
2.4	**Column C-4**										
	Dowel Bars 10'6" +12(0.5)" =11'	16	4	0.5		11	0.5	11.5	0.67	122.67	0.06

	Ties up to roof (10'6"-6") =10'@8 (Span/Spacing)+1 =16 levels	32	3	0.375		4	0.5	4.5	0.38	54.00	0.02
	Total R/f			0				0	0.00	**2136.19**	**0.97**
5	**R/f in Slabs**										
	Roof Slab										
5.1	**Two Side Continous Slab**										
	Bathroom and Dressing (13x6)										
	Long Side bar (6.5+0.5-0.083-0.083) =6.834@8"c/c (Span/Spacing)+1 =12 Bar	12	3	0.375		13.834	5.381	19.215	0.38	86.47	0.04
	Short Side Bar (13.5+0.5-0.083-0.083) =13.384@8"c/c (Span/Spacing)+1 =21 Bar	21	3	0.375		6.834	3.631	10.465	0.38	82.41	0.04
	Top Extra Long Side Bar (6.5+0.5-0.083-0.083) =6.834@8"c/c (Span/Spacing)+1 =12 Bar	12	3	0.375		3.375	0	3.375	0.38	15.19	0.01
	Top Extra Short Side Bar (13.5+0.5-0.083-0.083) =13.384@8"c/c (Span/Spacing)+1 =21 Bar	21	3	0.375		1.625	0	1.625	0.38	12.80	0.01
	Bathroom and Dressing (13x6)									196.86	0.09
	Total R/f per sqft									2.52	0.00
	Bath and Dressing (13.5x6)									204.44	0.09
	Drawing (15.5x16)									625.93	0.28
	Powder (7.167x4)									72.35499981	0.03
	Bath (5x6)									75.71682692	0.03
	Total R/f in Two Sided Slabs									1175.30	0.53

5.1	**Three Side Continous Slab**										
	Bed Room (15x13)										
	Long Side bar (6.5+0.5-0.083-0.083) =6.834@8"c/c (Span/Spacing)+1 =12 Bar	12	3	0.375		15.834	3.262	19.096	0.38	85.93	0.04
	Short Side bar (15.5+0.5-0.083-0.083) =15.834@8"c/c (Span/Spacing)+1 =25	25	3	0.375		6.834	3.631	10.465	0.38	98.11	0.04
	Top Extra Long Side Bar (6.5+0.5-0.083-0.083) =6.834@8"c/c (Span/Spacing)+1 =12 Bar	12	3	0.375		7.75	0	7.75	0.38	34.88	0.02
	Top Extra Short Side bar (15.5+0.5-0.083-0.083) =15.834@8"c/c (Span/Spacing)+1 =25	21	3	0.375		1.625	0	1.625	0.38	12.80	0.01
	Bed Room (15x13)									231.71	0.11
	Total R/f per sqft									1.19	0.00
	Kitchen (12x8)									114.07	0.05
	Kitchen (10.75x13.5)									172.45	0.08
	Entrance Lobby (14.5x9.5)									163.68	0.07
	Lounge/Dinning (22x17)									444.41	0.20
	Bath and Dressing (6x16)									114.07	0.05
	Bed Room (16x13)									114.07	0.05
	Bed Room (14x15)									2.92	0.00
	Total R/f in Three Sided Slabs									1357.41	0.62
5.1	**Four Side Continous Slab**										
	Passage (5x20)										
	Long Side bar (5.5+0.5-0.083-0.083) =5.834@8"c/c (Span/Spacing)+1 =10 Bar	10	3	0.375		20.334	3.262	23.596	0.38	88.49	0.04
	Short Side bar (20+0.5-0.083-0.083) =20.334@8"c/c (Span/Spacing)+1 =32 Bars	32	3	0.375		5.834	3.262	9.096	0.38	109.15	0.05

	Top Extra Long Side bar (5.5+0.5-0.083-0.083) =5.834@8"c/c (Span/Spacing)+1 =10 Bar	12	3	0.375		10	0	10	0.38	45.00	0.02
	Top Extra Short Side bar (20+0.5-0.083-0.083) =20.334@8"c/c (Span/Spacing)+1 =32 Bars	21	3	0.375		2.75	0	2.75	0.38	21.66	0.01
	Total R/f in Four Sided Slabs									264.29	0.12
	Total R/f in Roof Slab									**2797.00**	**1.27**
	First Floor Slab (Ground floor R/f + Terrace Slab)										
	Terrace (Two Side Continous 11.5x14.5) =R/f per sqft * Area									420.86	0.191
	Total R/f in First floor Slab									**3217.86**	**1.460**
	Total Reinforcement in Slab									**6014.85**	**2.73**
	Staircase Reinforcement Calculation (MKS)										
	Waist Slab 01	**Nos**	**Dia**	**Dia**	**Shape**	**SL**	**HL**	**TL**	**D^2/162**	**Kg**	**Tonnes**
	X Bar 8mm @100mm	12	8	8		4.6	0.072	4.672	0.39	21.84	0.02
	Y bar 8mm @100mm	34	8	8		1.2	0.8	2	0.39	30.42	0.03
	Waist Slab 02										
	X Bar 8mm @100mm	12	8	8		4.5	1.15	5.65	0.39	26.44	0.03
	Y bar 8mm @100mm	34	8	8		1.2	0.8	2	0.39	30.42	0.03
	Waist Slab 03										
	X Bar 8mm @100mm	12	8	8		4.6	0.072	4.672	0.39	21.84	0.02
	Y bar 8mm @100mm	34	8	8		1.2	0.8	2	0.39	30.42	0.03
	Waist Slab 04										
	X Bar 8mm @100mm	12	8	8		4.6	0.072	4.672	0.39	21.84	0.02
	Y bar 8mm @100mm	34	8	8		1.2	0.8	2	0.39	30.42	0.03
	Landing 01										
	Y Bar 8mm @100mm	12	8	8		2.5	0	2.5	0.39	11.23	0.01
	Landing 02										
	Y Bar 8mm @100mm	12	8	8		2.5	0	2.5	0.39	11.23	0.01
	L Type Bars for Stairs	40	8	8		1.8	1.316	3.116	0.39	48.61	0.05
	Total Reinforcement in Staircase									**333.3192**	**0.3333192**

MS (Measurement Sheet)

S NO	Description Of Works	Numbers	Length (ft)	Breadth (ft)	Height (ft)	Quantity (sft)	Total Quantity (sft)
			Quantities for Plaster				
1	**BOUNDARY WALLS**						
1.1	**LONG WALLS**						
	Longer side 01 (Left)	1	88.75	-	8	710	710
	Longer side 01 (Right)	1	88.75	-	8	710	710
1.2	**SHORT WALLS**						
	Shorter side 01 (Back)	1	50	-	8	400	400
	Shorter side 02 (front)	1	50	-	8	400	400
						Total	**2220**
1.3	**DEDUCTIONS OF BOUNDARY WALL**						
	Sliding Door	1	13.25	-	8	106	106
	Main Door	1	4.25	-	8	34	34
						Total	**140**
	Total Boundary Wall Plaster Area = 2220-140 = 2080 sft						
			GROUND FLOOR				
2	**EXTERNAL PLASTER**						
2.1	**WALLS**						
	Long Walls	2	68.5	-	10	685	1370
	Short Walls	2	36	-	10	360	720
						Total	**2090**
2.2	**SUNSHADES**						
2.2.1	**Long Wall 01 (left)**						
	for long side of W1	2	6	2	-	12	24
	for long side of D1	2	6	2	-	12	24
	for long side of W2	2	5.667	2	-	11.334	22.668
	for long side of D2	2	4.5	2	-	9	18
	for long side of D3	2	6	2	-	12	24
	for long side of W3	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of D1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
	for short side of D2	2	2	-	0.5	1	2
	for short side of D3	2	2	-	0.5	1	2
	for short side of W3	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
						Total	**182.668**
2.2.2	**Long Wall 01 (Right)**						
	for long side of W1	2	12	2	-	24	48
	for long side of W2	2	8.08	2	-	16.16	32.32
	for long side of W3	2	6.833	2	-	13.666	27.332
	for long side of W4	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
	for short side of W3	2	2	-	0.5	1	2
	for short side of W4	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
						Total	**191.652**
2.2.3	**Short Wall 01 (Front)**						
	for long side of W1	2	12.5	2	-	25	50
	for long side of V1	2	4	2	-	8	16
	for long side of D1	2	6.5	2	-	13	26
	for long side of W2	2	6	2	-	12	24
	for short side of W1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
	for short side of D1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
						Total	**124**
2.2.4	**Short Wall 02 (Back)**						
	for long side of W1	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
						Total	**60**
	Total Area of Sunshades for Plaster = 182.668+191.652+124+60 =558.32 sft						

2.3	DEDUCTIONS						
2.3.1	**Long Wall 01 (Left)**						
	W1	1	4	-	4	16	16
	D1	1	4	-	7	28	28
	W2	1	3.667	-	4	14.668	14.668
	D2	1	2.5	-	7	17.5	17.5
	D3	1	4	-	7	28	28
	W3	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
						Total	**139.168**
2.3.2	**Long Wall 02 (Right)**						
	W1	1	10	-	4	40	40
	W2	1	6.083	-	4	24.332	24.332
	W3	1	4.833	-	4	19.332	19.332
	W4	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
	V1	1	2	-	1.5	3	3
						Total	**121.664**
2.3.3	**Short Wall 01 (Front)**						
	W1	1	10.5	-	4	42	42
	V1	1	2	-	1.5	3	3
	D1	1	4.5	-	7	31.5	31.5
	W2	1	4	-	4	16	16
						Total	**92.5**
2.3.4	**Short Wall 02 (Back)**						
	W1	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
						Total	**35**
	Total Deductions = 139.168+121.664+92.5+35 = 388.332 sft						
	Total External Plaster Area of Ground Floor = 2090+558.32-388.332 = 2259.988 sft						
3	**INTERNAL PLASTER**						
3.1	**Drawing Room**						
	Ceiling	1	16	15.5	-	248	248
	Walls	1	63	-	10	630	630
3.2	**Powder Room**						
	Ceiling	1	7.1667	4	-	28.6668	28.6668
	Walls	1	22.33	-	10	223.3	223.3
3.3	**Lounge/Dinning**						
	Ceiling	1	22	17	-	374	374
	Walls	1	78	-	10	780	780
3.4	**Bedroom 01**						
	Ceiling	1	16	13	-	208	208
	Walls	1	58	-	10	580	580
3.5	**Bath 01**						
	Ceiling	1	10	6	-	60	60
	Walls	1	32	-	10	320	320
3.6	**Dress 01**						
	Ceiling	1	6	5.5	-	33	33
	Walls	1	23	-	10	230	230
3.5	**Bedroom 02**						
	Ceiling	1	15	14	-	210	210
	Walls	1	58	-	10	580	580
3.6	**Bath 02**						
	Ceiling	1	9	6	-	54	54
	Walls	1	30	-	10	300	300
3.7	**Dress 02**						
	Ceiling	1	6	4	-	24	24
	Walls	1	20	-	10	200	200
3.8	**Bedroom 03**						
	Ceiling	1	15	13	-	195	195
	Walls	1	56	-	10	560	560
3.9	**Bath 03**						
	Ceiling	1	8.5	6	-	51	51
	Walls	1	29	-	10	290	290
3.1	**Dress 03**						
	Ceiling	1	6	4	-	24	24
	Walls	1	20	-	10	200	200
3.11	**D Kitchen**						

	Ceiling	1	12	8	-	96	96
	Walls	1	40	-	10	400	400
3.12	**Bath (Adjacent with D Kitchen)**						
	Ceiling	1	6	5	-	30	30
	Walls	1	22	-	10	220	220
3.13	**Kitchen**						
	Ceiling	1	13.5	10.75	-	145.125	145.125
	Walls	1	48.5	-	10	485	485
3.14	**Entrance/Lobby**						
	Ceiling	1	14.5	9.5	-	137.75	137.75
	Walls	1	48	-	10	480	480
3.15	**Stairs**						
	Ceiling	1	14.25	9.334	-	133.0095	133.0095
	Walls	1	47.167	-	10	471.67	471.67
3.16	**Passage Area**						
	Ceiling	1	19	5	-	95	95
	Walls	1	48	-	10	480	480
						TOTAL	**9576.5213**
4	**JAMB**						
4.1	**Entrance/Lobby**						
	D1	1	18.5	0.5	-	9.25	9.25
	D2	1	18.5	0.5	-	9.25	9.25
	D3	1	19	0.5	-	9.5	9.5
	D4	1	17.5	0.5	-	8.75	8.75
4.2	**Drawing Room**						
	W1	1	29	0.5	-	14.5	14.5
	W2	1	28	0.5	-	14	14
	Sliding Door	1	24.5	0.5	-	12.25	12.25
4.3	**Lounge/Dinning**						
	W1	1	20.1666	0.5	-	10.0833	10.0833
	W2	1	29.666	0.5	-	14.833	14.833
4.4	**Bedroom 01**						
	W1	1	24	0.5	-	12	12
	D1	1	17.75	0.5	-	8.875	8.875
4.5	**Bath 01**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.6	**Bedroom 02**						
	W1	1	24	0.5	-	12	12
	D1	1	17.5	0.5	-	8.75	8.75
4.7	**Bath 02**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.8	**Bedroom 03**						
	W1	1	24	0.5	-	12	12
	D1	1	17.5	0.5	-	8.75	8.75
4.9	**Bath 03**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.1	**D Kitchen**						
	D1	1	16.5	0.5	-	8.25	8.25
	W1	1	23	0.5	-	11.5	11.5
4.11	**Bath (Adjecent to D kitchen)**						
	D1	1	16.5	0.5	-	8.25	8.25
4.12	**Kitchen**						
	W1	1	23	0.5	-	11.5	11.5
4.13	**Stairs**						
	D1	1	16.5	0.5	-	8.25	8.25
	W1	1	15	0.5	-	7.5	7.5
	W2	1	16	0.5	-	8	8
4.14	**Powder Room**						
	V1	1	8	0.5	-	4	4
	D1	1	16.5	0.5	-	8.25	8.25
						TOTAL	**277.0413**
5	**DEDUCTIONS**						
5.1	**Entrance/Lobby**						
	D1	1	4.5	-	7	31.5	31.5
	D2	1	4.5	-	7	31.5	31.5
	D3	1	5	-	7	35	35
	D4	1	3.5	-	7	24.5	24.5
5.2	**Drawing Room**						
	W1	1	10.5	-	4	42	42

	W2	1	10	-	4	40	40
	Sliding Door	1	10.5	-	7	73.5	73.5
5.3	**Lounge/Dinning**						
	W1	1	6.0833	-	4	24.3332	24.3332
	W2	1	4.833	-	4	19.332	19.332
5.4	**Bedroom 01**						
	W1	1	8	-	4	32	32
	D1	1	3.75	-	7	26.25	26.25
5.5	**Bath 01**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.6	**Bedroom 02**						
	W1	1	8	-	4	32	32
	D1	1	3.5	-	7	24.5	24.5
5.7	**Bath 02**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.8	**Bedroom 03**						
	W1	1	8	-	4	32	32
	D1	1	3.5	-	7	24.5	24.5
5.9	**Bath 03**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.1	**D Kitchen**						
	D1	1	2.5	-	7	17.5	17.5
	W1	1	3.5	-	4	14	14
5.11	**Bath (Adjecent to D kitchen)**						
	D1	1	2.5	-	7	17.5	17.5
5.12	**Kitchen**						
	W1	1	3.5	-	4	14	14
5.13	**Stairs**						
	D1	1	3.5	-	7	24.5	24.5
	W1	1	4	-	4	16	16
	W2	1	4	-	4	16	16
5.14	**Powder Room**						
	V1	1	2	-	1.5	3	3
	D1	1	2.5	-	7	17.5	17.5
						TOTAL	**694.4152**
	Total Internal Plaster Area of Ground Floor = 9576.5213+277.0413-694.4152=9159.14sft						
	TOTAL PLASTER OF GROUND FLOOR						
		Total Plaster = External plaster + Internal platser + Boundary Wall					
		Total Plaster = 2259.988+9159.14+2080 = 13499.128 sft					
		FIRST FLOOR					
2	**EXTERNAL PLASTER**						
2.1	**WALLS**						
	Long Walls	2	68.5	-	10	685	1370
	Short Walls	2	36	-	10	360	720
						Total	**2090**
2.2	**SUNSHADES**						
2.2.1	**Long Wall 01 (left)**						
	for long side of W1	2	6	2		12	24
	for long side of W2	2	5.667	2	-	11.334	22.668
	for long side of W3	2	3	2		6	12
	for long side of D1 (Servant Room)	2	4.5	2	-	9	18
	for long side of D2 (Servant Room)	2	4.5	2	-	9	18
	for long side of W4	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
	for short side of W3	2	2	-	0.5	1	2
	for short side of D1	2	2	-	0.5	1	2
	for short side of D2	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
	for short side of W4	2	2	-	0.5	1	2
						Total	**164.668**
2.2.2	**Long Wall 01 (Right)**						
	for long side of W1	2	12	2	-	24	48
	for long side of W2	2	8.08	2	-	16.16	32.32

	for long side of W3	2	6.833	2	-	13.666	27.332
	for long side of W4	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
	for short side of W3	2	2	-	0.5	1	2
	for short side of W4	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
						Total	**191.652**
2.2.3	**Short Wall 01 (Front)**						
	for long side of W1	2	11.5	2	-	23	46
	for long side of V1	2	4	2	-	8	16
	for long side of D1	2	5.5	2	-	11	22
	for long side of W2	2	6	2	-	12	24
	for short side of W1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
	for short side of D1	2	2	-	0.5	1	2
	for short side of W2	2	2	-	0.5	1	2
						Total	**116**
2.2.4	**Short Wall 02 (Back)**						
	for long side of W1	2	10	2	-	20	40
	for long side of V1	2	4	2	-	8	16
	for short side of W1	2	2	-	0.5	1	2
	for short side of V1	2	2	-	0.5	1	2
						Total	**60**
	Total Area of Sunshades for Plaster = 164.668+191.652+116+60 =532.32 sft						
2.3	**DEDUCTIONS**						
2.3.1	**Long Wall 01 (Left)**						
	W1	1	4	-	4	16	16
	W2	1	3	-	4	12	12
	W2	1	3.667	-	4	14.668	14.668
	D2	1	2.5	-	7	17.5	17.5
	D3	1	4	-	7	28	28
	W3	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
						Total	**123.168**
2.3.2	**Long Wall 02 (Right)**						
	W1	1	10	-	4	40	40
	W2	1	6.083	-	4	24.332	24.332
	W3	1	4.833	-	4	19.332	19.332
	W4	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
	V1	1	2	-	1.5	3	3
						Total	**121.664**
2.3.3	**Short Wall 01 (Front)**						
	W1	1	9.5	-	4	38	38
	V1	1	2	-	1.5	3	3
	D1	1	3.5	-	7	24.5	24.5
	W2	1	4	-	4	16	16
						Total	**81.5**
2.3.4	**Short Wall 02 (Back)**						
	W1	1	8	-	4	32	32
	V1	1	2	-	1.5	3	3
						Total	**35**
	Total Deductions = 123.168+121.664+81.5+35 = 361.332 sft						
	Total External Plaster Area of First Floor = 2090+532.35-361.332 = 2261.018 sft						
3	**INTERNAL PLASTER**						
3.1	**Drawing Room**						
	Ceiling	1	15.6	14	-	218.4	218.4
	Walls	1	63	-	10	630	630
3.2	**Powder Room**						
	Ceiling	1	7.025	4	-	28.1	28.1
	Walls	1	22.05	-	10	220.5	220.5
3.3	**Lounge/Dinning**						
	Ceiling	1	22	17	-	374	374

	Walls	1	78	-	10	780	780
3.4	**Bedroom 01**						
	Ceiling	1	16	13	-	208	208
	Walls	1	58	-	10	580	580
3.5	**Bath 01**						
	Ceiling	1	10	6	-	60	60
	Walls	1	32	-	10	320	320
3.6	**Dress 01**						
	Ceiling	1	6	5.5	-	33	33
	Walls	1	23	-	10	230	230
3.5	**Bedroom 02**						
	Ceiling	1	15	14	-	210	210
	Walls	1	58	-	10	580	580
3.6	**Bath 02**						
	Ceiling	1	9	6	-	54	54
	Walls	1	30	-	10	300	300
3.7	**Dress 02**						
	Ceiling	1	6	4	-	24	24
	Walls	1	20	-	10	200	200
3.8	**Bedroom 03**						
	Ceiling	1	15	13	-	195	195
	Walls	1	56	-	10	560	560
3.9	**Bath 03**						
	Ceiling	1	8.5	6	-	51	51
	Walls	1	29	-	10	290	290
3.1	**Dress 03**						
	Ceiling	1	6	4	-	24	24
	Walls	1	20	-	10	200	200
3.11	**Servant Room 01**						
	Ceiling	1	8	6	-	48	48
	Walls	1	28	-	10	280	280
3.12	**Servant Room 02**						
	Ceiling	1	8	6	-	48	48
	Walls	1	28	-	10	280	280
3.13	**Kitchen**						
	Ceiling	1	13.5	10.75	-	145.125	145.125
	Walls	1	48.5	-	10	485	485
3.14	**Entrance/Lobby**						
	Ceiling	1	14.5	9.5	-	137.75	137.75
	Walls	1	48	-	10	480	480
3.15	**Stairs**						
	Ceiling	1	14.25	9.334	-	133.0095	133.0095
	Walls	1	47.167	-	10	471.67	471.67
3.16	**Passage Area**						
	Ceiling	1	19	5	-	95	95
	Walls	1	48	-	10	480	480
						TOTAL	**9453.5545**
4	**JAMB**						
4.1	**Entrance/Lobby**						
	D1	1	17.5	0.5	-	8.75	8.75
	D2	1	18.916	0.5	-	9.458	9.458
	D3	1	19	0.5	-	9.5	9.5
	D4	1	17.5	0.5	-	8.75	8.75
4.2	**Drawing Room**						
	W1	1	27	0.5	-	13.5	13.5
	W2	1	28	0.5	-	14	14
	Sliding Door	1	23.5	0.5	-	11.75	11.75
4.3	**Lounge/Dinning**						
	W1	1	20.1666	0.5	-	10.0833	10.0833
	W2	1	29.666	0.5	-	14.833	14.833
4.4	**Bedroom 01**						
	W1	1	24	0.5	-	12	12
	D1	1	17.5	0.5	-	8.75	8.75
4.5	**Bath 01**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.6	**Bedroom 02**						
	W1	1	24	0.5	-	12	12
	D1	1	17.5	0.5	-	8.75	8.75
4.7	**Bath 02**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.8	**Bedroom 03**						
	W1	1	24	0.5	-	12	12

	D1	1	17.5	0.5	-	8.75	8.75
4.9	**Bath 03**						
	D1	1	16.5	0.5	-	8.25	8.25
	V1	1	8	0.5	-	4	4
4.1	**Servant Room 01**						
	W1	1	11.334	0.5	-	5.667	5.667
	D1	1	16.5	0.5	-	8.25	8.25
4.11	**Servant Room 02**						
	W1	1	11.334	0.5	-	5.667	5.667
	D1	1	16.5	0.5	-	8.25	8.25
4.12	**Kitchen**						
	W1	1	14	0.5	-	7	7
4.13	**Stairs**						
	W1	1	15	0.5	-	7.5	7.5
	W2	1	16	0.5	-	8	8
4.14	**Powder Room**						
	V1	1	8	0.5	-	4	4
	D1	1	16.5	0.5	-	8.25	8.25
						TOTAL	**246.9583**
5	**DEDUCTIONS**						
5.1	**Entrance/Lobby**						
	D1	1	3.5	-	7	24.5	24.5
	D2	1	4.916	-	7	34.412	34.412
	D3	1	5	-	7	35	35
	D4	1	3.5	-	7	24.5	24.5
5.2	**Drawing Room**						
	W1	1	9.5	-	4	38	38
	W2	1	10	-	4	40	40
	Sliding Door	1	9.5	-	7	66.5	66.5
5.3	**Lounge/Dinning**						
	W1	1	6.0833	-	4	24.3332	24.3332
	W2	1	4.833	-	4	19.332	19.332
5.4	**Bedroom 01**						
	W1	1	8	-	4	32	32
	D1	1	3.5	-	7	24.5	24.5
5.5	**Bath 01**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.6	**Bedroom 02**						
	W1	1	8	-	4	32	32
	D1	1	3.5	-	7	24.5	24.5
5.7	**Bath 02**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.8	**Bedroom 03**						
	W1	1	8	-	4	32	32
	D1	1	3.5	-	7	24.5	24.5
5.9	**Bath 03**						
	D1	1	2.5	-	7	17.5	17.5
	V1	1	2	-	1.5	3	3
5.1	**Servant Room 01**						
	W1	1	1.667	-	4	6.668	6.668
	D1	1	2.5	-	7	17.5	17.5
5.11	**Servant Room 02**						
	D1	1	2.5	-	7	17.5	17.5
	W1	1	1.667	-	4	6.668	6.668
5.12	**Kitchen**						
	W1	1	3	-	4	12	12
5.13	**Stairs**						
	W1	1	4	-	4	16	16
	W2	1	4	-	4	16	16
5.14	**Powder Room**						
	V1	1	2	-	1.5	3	3
	D1	1	2.5	-	7	17.5	17.5
						TOTAL	**631.7452**

Total Internal Plaster Area of First Floor = 9453.5545+246.9583-631.7452=9068.7676sft

TOTAL PLASTER OF GROUND FLOOR

Total Plaster = External plaster + Internal platser

Total Plaster = 2261.018+9068.7676 = 11329.7856 sft

TOTAL PLASTER

Total Plaster = Total Plaster Area Of Boundary Walls +Total Plaster Area Of External Walls + Total Plaster Area Of Internal Walls

Total Plaster = 2080+13499.128+11329.78856 = 26908.9136 sft

Number of bag = (Area of Plaster x Thickness of Plaster (0.75" thick) x Plaster Ratio (1:6) x Wastage) / Volume of 1 Cement Bag

Number of bag = (26908.9136 x (0.75/12) x (1/7) x 1.2) / 1.25

Number of bag = 231 bags (@ 1200 Rupees/Bag)

Total Amount of bags = 2,77,200 Rupees

MEASUREMENT SHEET (MS)							
S NO	Description Of Work	Number	Length (ft)	Breadth (ft)	Height (ft)	Quantity (cft)	Total Quantity (cft)
			Flooring And Marble				
GROUND FLOOR							
1	**Quantities For Marble Flooring**						
1.1	**Drawing Room**						
	Floor	1	15.5	16	0	248	248
	Skirting (4" depth)	0	63	0	0.333	20.979	20.979
1.2	**Powder Room**						
	Floor	1	7.1667	4	0	28.6668	28.6668
	Skirting (4" depth)	0	22.3334	0	0.333	7.4370222	7.4370222
1.3	**Lounge/Dinning**						
	Floor	1	22	17	0	374	374
	Skirting (4" depth)	0	78	0	0.333	25.974	25.974
1.4	**Bedroom 1**						
	Floor	1	16	13	0	208	208
	Skirting (4" depth)	0	58	0	0.333	19.314	19.314
1.5	**Dress 01**						
	Floor	1	6	5.5	0	33	33
	Skirting (4" depth)	0	23	0	0.333	7.659	7.659
1.6	**Bedroom 2**						
	Floor	1	14	15	0	210	210
	Skirting (4" depth)	0	58	0	0.333	19.314	19.314
1.7	**Dress 02**						
	Floor	1	4	6	0	24	24
	Skirting (4" depth)	0	20	0	0.333	6.66	6.66
1.8	**Bedroom 03**						
	Floor	1	15	13	0	195	195
	Skirting (4" depth)	0	56	0	0.333	18.648	18.648
1.9	**Dress 03**						
	Floor	1	6	4	0	24	24
	Skirting (4" depth)	0	20	0	0.333	6.66	6.66
1.1	**D Kitchen**						
	Floor	1	12	8	0	96	96
	Skirting (4" depth)	0	40	0	0.333	13.32	13.32
1.11	**Kitchen**						
	Floor	1	10.75	13.5	0	145.125	145.125
	Skirting (4" depth)	0	48.5	0	0.333	16.1505	16.1505
1.12	**Entrance Lobby**						
	Floor	1	14.5	9.5	0	137.75	137.75
	Skirting (4" depth)	0	48	0	0.333	15.984	15.984
1.13	**Passage Area**						
	Floor	1	19	5	0	95	95
	Skirting (4" depth)	0	48	0	0.333	15.984	15.984
1.14	**Parking Area**						
	Floor	1	21.583	21.5	0	464.0345	464.0345
	Skirting (4" depth)	0	86.166	0	0.333	28.693278	28.693278
1.15	**Door's Bottom**						
	D1	2	4.5	0.5	0	4.5	4.5
	D2	5	3.5	0.5	0	8.75	8.75
	D3 and D4	6	2.5	0.5	0	7.5	7.5
	D5	1	5	0.5	0	2.5	2.5
1.16	**Deductions for Marble Skirting**						
	D1	4	4.5	0	0.333	5.994	5.994
	D2	10	3.5	0	0.333	11.655	11.655
	D3 and D4	7	2.5	0	0.333	5.8275	5.8275
	D5	2	5	0	0.333	3.33	3.33
1.17	**Steps**						
	Step	17	3.583	1	0	60.911	60.911
	Risers	34	3.583	0	0.5	60.911	60.911
	Total Floor Marble Area:	**2305.8263**	**sft**				
	Skirting Area:	**222.7768002**	**sft**				
	Total Skirting Area:	**195.9703002**	**sft**				
	Steps Area:	**121.822**	**sft**				
2	**Quantities for Bathroom Tiles for 8" x 8" Tiles**						
2.1	**Bathroom 01**						
	Floor Area	1	10	6	0	60	60
	Long Wall	2	10	0	5	100	100
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**207.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	471.5909091					
	No of tiles =	**472**	**Tiles Approx.**				
2.2	**Bathroom 02**						
	Floor Area	1	9	6	0	54	54
	Long Wall	2	9	0	5	90	90
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**191.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	435.2272727					
	No of tiles =	**436**	**Tiles Approx.**				
2.3	**Bathroom 03**						
	Floor Area	1	8.5	6	0	51	51

	Long Wall	2	8.5	0	5	85	85
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**183.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	417.0454545					
	No of tiles =	**418**	**Tiles Approx.**				
2.4	**Bathroom 04**						
	Floor Area	1	6	5	0	30	30
	Long Wall	2	6	0	5	60	60
	Short Wall	2	5	0	5	50	50
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**127.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	289.7727273					
	No of tiles =	**290**	**Tiles Approx.**				
FIRST FLOOR							
1	**Quantities For Marble Flooring**						
1.1	**Drawing Room**						
	Floor	1	15.5	14	0	217	217
	Skirting (4” depth)	0	59	0	0.333	19.647	19.647
1.2	**Powder Room**						
	Floor	1	6.25	4	0	25	25
	Skirting (4” depth)	0	20.5	0	0.333	6.8265	6.8265
1.3	**Lounge/Dinning**						
	Floor	1	22	17	0	374	374
	Skirting (4” depth)	0	78	0	0.333	25.974	25.974
1.4	**Bedroom 1**						
	Floor	1	16	13	0	208	208
	Skirting (4” depth)	0	58	0	0.333	19.314	19.314
1.5	**Dress 01**						
	Floor	1	6	5.5	0	33	33
	Skirting (4” depth)	0	23	0	0.333	7.659	7.659
1.6	**Bedroom 2**						
	Floor	1	14	15	0	210	210
	Skirting (4” depth)	0	58	0	0.333	19.314	19.314
1.7	**Dress 02**						
	Floor	1	4	6	0	24	24
	Skirting (4” depth)	0	20	0	0.333	6.66	6.66
1.8	**Bedroom 03**						
	Floor	1	15	13	0	195	195
	Skirting (4” depth)	0	56	0	0.333	18.648	18.648
1.9	**Dress 03**						
	Floor	1	6	4	0	24	24
	Skirting (4” depth)	0	20	0	0.333	6.66	6.66
1.1	**Servant**						
	Floor	2	8	6	0	96	96
	Skirting (4” depth)	0	56	0	0.333	18.648	18.648
1.11	**Room with Servant**						
	Floor	1	6	5	0	30	30
	Skirting (4” depth)		22	0	0.333	7.326	7.326
1.12	**Kitchen**						
	Floor	1	10.75	13.5	0	145.125	145.125
	Skirting (4” depth)	0	48.5	0	0.333	16.1505	16.1505
1.13	**Entrance Lobby**						
	Floor	1	14.5	9.5	0	137.75	137.75
	Skirting (4” depth)	0	48	0	0.333	15.984	15.984
1.14	**Passage Area**						
	Floor	1	19	5	0	95	95
	Skirting (4” depth)	0	48	0	0.333	15.984	15.984
1.15	**Terrace**						
	Floor	1	14.5	11.5	0	166.75	166.75
	Skirting (4” depth)	0	52	0	0.333	17.316	17.316
1.16	**Door's Bottom**						
	D1	1	4.5	0.5	0	2.25	2.25
	D2	5	3.5	0.5	0	8.75	8.75
	D3 and D4	6	2.5	0.5	0	7.5	7.5
	D5	1	5	0.5	0	2.5	2.5
1.17	**Deductions for Marble Skirting**						
	D1	2	4.5	0	0.333	2.997	2.997
	D2	10	3.5	0	0.333	11.655	11.655
	D3 and D4	9	2.5	0	0.333	7.4925	7.4925
	D5	2	5	0	0.333	3.33	3.33
1.18	**Steps**						
	Step	17	3.583	1	0	60.911	60.911
	Risers	34	3.583	0	0.5	60.911	60.911
	Total Floor Marble Area:	**2001.625**	**sft**				
	Skirting Area:	**222.111**	**sft**				
	Total Skirting Area:	**196.6365**	**sft**				
	Steps Area:	**121.822**	**sft**				
2	**Quantities for Bathroom Tiles for 8” x 8” Tiles**						
2.1	**Bathroom 01**						
	Floor Area	1	10	6	0	60	60
	Long Wall	2	10	0	5	100	100
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**207.5**	**sft**				

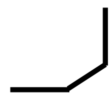
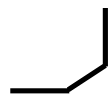
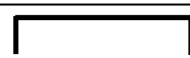
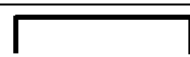
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	471.5909091					
	No of tiles =	**472**	**Tiles Approx.**				
2.2	**Bathroom 02**						
	Floor Area	1	9	6	0	54	54
	Long Wall	2	9	0	5	90	90
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**191.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	435.2272727					
	No of tiles =	**436**	**Tiles Approx.**				
2.3	**Bathroom 03**						
	Floor Area	1	8.5	6	0	51	51
	Long Wall	2	8.5	0	5	85	85
	Short Wall	2	6	0	5	60	60
	Deduction For D4	1	2.5	0	5	12.5	12.5
	Total Tiles Area	**183.5**	**sft**				
	No of tiles = Floor Area/ Tile Area						
	No of tiles =	417.0454545					
	No of tiles =	**418**	**Tiles Approx.**				

MEASUREMENT SHEET (MS)							
S NO	**Description Of Work**	**Number**	**Length (ft)**	**Breadth (ft)**	**Height (ft)**	**Quantity (cft)**	**Total Quantity (cft)**
				Block Masonary			
	GROUND FLOOR						
1	**Long Walls**						
	Grid A (with offset wall)	1	50.58	0.50	8.00	202.33	202.33
	Grid B-1	1	4.83	0.50	8.00	19.33	19.33
	Grid C	1	5.83	0.50	8.00	23.33	23.33
	Grid D	1	14.00	0.50	8.00	56.00	56.00
	Grid F	1	41.00	0.50	8.00	164.00	164.00
	Grid G	1	7.17	0.50	8.00	28.67	28.67
	Grid H	1	17.00	0.50	8.00	68.00	68.00
	Grid J	1	14.50	0.50	8.00	58.00	58.00
	Grid K (offset wall)	1	7.00	0.50	8.00	28.00	28.00
	Grid M	1	13.50	0.50	8.00	54.00	54.00
	Grid P	1	60.67	0.50	8.00	242.67	242.67
2	**Short Walls**						
	Grid 1	1	31.50	0.50	8.00	126.00	126.00
	Grid 2	1	13.00	0.50	8.00	52.00	52.00
	Grid 3-A (Offset wall)	1	5.00	0.50	8.00	20.00	20.00
	Grid 3	1	18.25	0.50	8.00	73.00	73.00
	Grid 4	1	11.50	0.50	8.00	46.00	46.00
	Grid 6-A	1	4.58	0.50	8.00	18.33	18.33
	Grid 6	1	13.50	0.50	8.00	54.00	54.00
	Grid 8	1	24.00	0.50	8.00	96.00	96.00
	Grid 9	1	9.50	0.50	8.00	38.00	38.00
	Grid 9 (Angled wall)	1	9.30	0.50	8.00	37.20	37.20
	Grid 10	1	14.00	0.50	8.00	56.00	56.00
	Grid 11	1	7.33	0.50	8.00	29.33	29.33
	Grid 12	1	8.17	0.50	8.00	32.67	32.67
	Grid 14	1	17.25	0.50	8.00	69.00	69.00
						Total	**1691.87**
3	**Deductions for openings**						
	D1	2	4.5	0.5	7	15.75	31.5
	D2	5	3.5	0.5	7	12.25	61.25
	D3 and D4	6	2.5	0.5	7	8.75	52.5
	D5	1	5	0.5	7	17.5	17.5
	D6 (Sliding)	1	10.5	0.5	7	36.75	36.75
	W1	3	8	0.5	4	16	48
	W2	2	4	0.5	4	8	16
	W3	1	10.5	0.5	4	21	21
	W4	1	10	0.5	4	20	20
	W5	1	6	0.5	4	12	12
	W6	1	4.8333	0.5	4	9.6666	9.6666
	W7	2	3.5	0.5	4	7	14
	V1	5	2.5	0.5	1	1.25	6.25
						Total	**346.4166**
	FIRST FLOOR						
1	**Long Walls**						
	Grid A (with offset wall)	1	50.583	0.5	8	202.332	202.332
	Grid B-1	1	4.833333333	0.5	8	19.33333333	19.33333333
	Grid C	1	5.833333333	0.5	8	23.33333333	23.33333333
	Grid D	1	14	0.5	8	56	56

	Grid F	1	41	0.5	8	164	164
	Grid G	1	7.1667	0.5	8	28.6668	28.6668
	Grid H	1	17	0.5	8	68	68
	Grid J	1	14.4997	0.5	8	57.9988	57.9988
	Grid K (offset wall)	1	7	0.5	8	28	28
	Grid M	1	13.5	0.5	8	54	54
	Grid P	1	60.6676	0.5	8	242.6704	242.6704
2	**Short Walls**						
	Grid 1	1	31.5	0.5	8	126	126
	Grid 2	1	13	0.5	8	52	52
	Grid 3-A (Offset wall)	1	5	0.5	8	20	20
	Grid 3	1	18.25	0.5	8	73	73
	Grid 4	1	11.5	0.5	8	46	46
	Grid 6-A	1	4.583333333	0.5	8	18.33333333	18.33333333
	Grid 6	1	13.5	0.5	8	54	54
	Grid 8	1	24	0.5	8	96	96
	Grid 9	1	9.5	0.5	8	38	38
	Grid 9 (Angled wall)	1	9.3	0.5	8	37.2	37.2
	Grid 10	1	14	0.5	8	56	56
	Grid 11	1	7.333	0.5	8	29.332	29.332
	Grid 12	1	8.167	0.5	8	32.668	32.668
	Grid 14	1	17.25	0.5	8	69	69
3	**Deductions for Openings**					**Total**	**1691.868**
	D1	6	2.5	0.5	7	8.75	52.5
	D2	5	3.5	0.5	7	12.25	61.25
	D3	2	5	0.5	7	17.5	35
	D4(Sliding)	1	9.5	0.5	7	33.25	33.25
	W1	5	8	0.5	4	16	80
	W2	1	6	0.5	4	12	12
	W3	1	4.8333	0.5	4	9.6666	9.6666
	W4	1	4	0.5	4	8	8
	V1	7	2	0.5	1	1	7
						Total	**298.6666**
	BOUNDARY WALL						
	Longer Side 01 (Left)	1	78.25	0.5	8	313	313
	Longer Side 02 (Right)	1	78.75	0.5	8	315	315
	Shorter Side 01 (Back)	1	45	0.5	8	180	180
	Shorter Side 02 (Front)	1	32	0.5	8	128	128
						Total	**936**
	Ground floor B.M =	**1345.45**	**cft**				
	First floor B.M =	**1393.2014**	**cft**				
	Boundary wall=	**936**	**cft**				
	Total Block Masonry	**3674.65**	**cft**				
	Number of Blocks = Volume of B.M*Mortar Deduction / Volume of 1 Block (6" x 9" x 12")						
	Number of Blocks =	**7839.2593**					
		7840	**Blocks(Approx)**				

BAR BENDING SCHEDULE (BBS)											
S NO	Description Of Work	Number	Bar Number	Diameter (in)	Shape	SL (ft)	HL (ft) (12Dia)	Total Length (ft)	Wt/ft	Weight (lbs)	Weight (tonnes)
Reinforcement in Tanks											
1	**R/f in UGWT**										
1.1	**Bottom Slab**										
	Bottom Main	13	3	0.375		6.0833	0.6667	6.75	0.38	32.91	0.01
	Bottom Distribution	9	3	0.375		9.0833	0.6667	9.75	0.38	32.91	0.01
	Top Main	13	3	0.375		6.0833	0.6667	6.75	0.38	32.91	0.01
	Top Distribution	9	3	0.375		9.0833	0.6667	9.75	0.38	32.91	0.01
	Corner Bars										
	Along Short Wall	8	3	0.375		6.0833		6.0833	0.38	18.25	0.01
	Along Long Wall	8	3	0.375		9.0833		9.0833	0.38	27.25	0.01
	Total R/f in bottom slab									**177.12**	**0.08**
1.2	**Top Slab**										
	Bottom Main	17	3	0.375		6.0833	0.75	6.8333	0.38	43.56	0.02
	Bottom Distribution	11	3	0.375		9.0833	0.75	9.8333	0.38	40.56	0.02
	Top Main	17	3	0.375		6.0833	0.75	6.8333	0.38	43.56	0.02
	Top Distribution	11	3	0.375		9.0833	0.75	9.8333	0.38	40.56	0.02
	Corner Bars										
	Along Short Wall	8	4	0.5		6.0833		6.0833	0.67	32.44	0.01
	Along Long Wall	8	4	0.5		9.0833		9.0833	0.67	48.44	0.02
	Total R/f in top slab									**249.14**	**0.11**

1.3	**Long Walls**										
	Inner Main	26	3	0.375		8.4167	0.667	9.0837	0.38	88.57	0.04
	Inner Distribution	20	3	0.375		9.0833	0.667	9.7503	0.38	73.13	0.03
	Outer Main	26	3	0.375		8.4167	1.9167	10.3334	0.38	100.75	0.05
	Outer Distribution	20	3	0.375		9.0833	0.667	9.7503	0.38	73.13	0.03
	Total R/f in long walls									**335.57**	**0.15**
1.4	**Short Walls**										
	Inner Main	19	3	0.375		8.4167	0.667	9.0837	0.38	64.72	0.03
	Inner Distribution	20	3	0.375		6.0833	0.667	6.7503	0.38	50.63	0.02

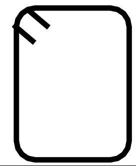
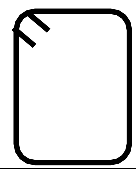
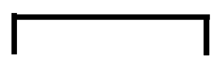
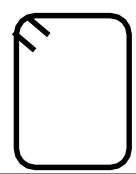
	Outer Main	19	3	0.375		8.4167	1.9167	10.3334	0.38	73.63	0.03
	Outer Distribution	20	3	0.375		6.0833	0.667	6.7503	0.38	50.63	0.02
	Total R/f in short walls									**239.60**	**0.11**
1.5	**Chamber R/f**										
	Along Short Wall	15	3	0.375		1.5	1.5	3	0.38	16.88	0.01
	Along Long Wall	21	3	0.375		1.5	1.5	3	0.38	23.63	0.01
	Total R/f in chamber									**40.50**	**0.02**
	Total Reinforcement in UGWT									**1041.94**	**0.47**
2	**R/f in OHWT**										
2.1	**Bottom Slab**										
	Bottom Main	13	3	0.375		7.1667	0.667	7.8337	0.38	38.19	0.02
	Bottom Distribution	14	3	0.375		7.1667	0.667	7.8337	0.38	41.13	0.02
	Top Main	13	3	0.375		7.1667	0.667	7.8337	0.38	38.19	0.02
	Top Distribution	15	3	0.375		7.1667	0.667	7.8337	0.38	44.06	0.02
	Total in R/f bottom slab									**161.57**	**0.07**
2.2	**Top Slab**										

	Bottom Main	9	3	0.375		7.1667	0.667	7.8337	0.38	26.44	0.01
	Bottom Distribution	9	3	0.375		7.1667	0.667	7.8337	0.38	26.44	0.01
	Top Main	11	3	0.375		7.1667		7.1667	0.38	29.56	0.01
	Total R/f in top slab									**82.44**	**0.04**
2.3	**Long Walls(including beams)**										
	Inner Main	20	3	0.375		8.54167	0.667	9.20867	0.38	69.07	0.03
	Inner Distribution	14	3	0.375		7.1667	0.667	7.8337	0.38	41.13	0.02
	Outer Main	20	3	0.375		8.54167	2.91667	11.45834	0.38	85.94	0.04
	Outer Distribution	14	3	0.375		7.1667	0.667	7.8337	0.38	41.13	0.02
	Total R/f in long walls									**237.26**	**0.11**
2.4	**Short Walls**										

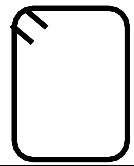
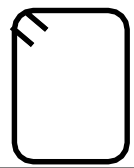
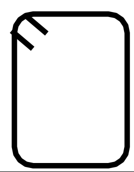
	Inner Main	20	3	0.375		5.79167	0.667	6.45867	0.38	48.44	0.02
	Inner Distribution	14	3	0.375		7.1667	0.667	7.8337	0.38	41.13	0.02
	Outer Main	20	3	0.375		5.79167	2.9167	8.70837	0.38	65.31	0.03
	Outer Distribution	14	3	0.375		7.1667	0.667	7.8337	0.38	41.13	0.02
	Total R/f in short walls									**196.01**	**0.09**
2.5	**Corner Bar**	16	4	0.5		7.1667		7.1667	0.67	**76.44**	**0.03**
Total Reinforcement in OHWT										**753.72**	**0.34**

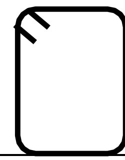
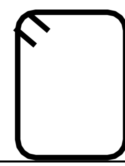
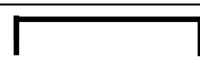
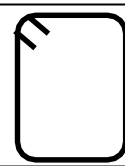
BAR BENDING SCHEDULE (BBS)											
S NO	Description Of Work	Number	Bar Number	Diameter (in)	Shape	SL (ft)	HL (ft) (12	Total Length (ft)	Wt/ft	Weight (lbs)	Weight (tonnes)
1	Reinforcement in Plinth Beams										
	Boundry Wall										
1.1	PB-01 6"x24" Short Wall upside										
	Bottom Through	2	4	0.5		49.75	1	50.75	0.333333333	33.83333333	0.015350877
	Bottom Extra	5	4	0.5		5		5	0.333333333	8.333333333	0.003781004
	Top Through	2	4	0.5		49.75	1	50.75	0.333333333	33.83333333	0.015350877
	Top Extra	4	4	0.5		5		5	0.333333333	6.666666667	0.003024803
	Stirrups	91	3	0.375		4.333	0.75	5.083	0.25	115.63825	0.052467446
	Total									198.3049167	0.089975008
1.2	PB-01(External Wall)6"x24" long Wall right side										
	Bottom Through	2	4	0.5		80.25	1	81.25	0.333333333	54.16666667	0.024576528
	Bottom Extra	9	4	0.5		7.369		7.369	0.333333333	22.107	0.010030399
	Top Through	2	4	0.5		80.25	1	81.25	0.333333333	54.16666667	0.024576528
	Top Extra	8	4	0.5		4.93		4.93	0.333333333	13.14666667	0.005964912
	Stirrups	160	3	0.375		4.333	0.75	5.083	0.25	203.32	0.092250454
	Total									346.907	0.15739882
1.3	PB-01(External Wall)6"x24" Short Wall Down side										
	Bottom Through	2	4	0.5		29.91667	1	30.91667	0.333333333	20.61111333	0.009351685

	Bottom Extra	3	4	0.5		7.479		7.479	0.333333333	7.479	0.003393376
	Top Through	2	4	0.5		29.91667	1	30.91667	0.333333333	20.61111333	0.009351685
	Top Extra	2	4	0.5		4.9861		4.9861	0.333333333	3.324066667	0.001508197
	Stirrups	55	3	0.375		4.333	0.75	5.083	0.25	69.89125	0.031711093
	Total									121.9165433	0.055316036
1.4	PB-01(External Wall)6"x24" long Wall left side										
	Bottom Through	2	4	0.5		78.833	1	79.833	0.333333333	53.222	0.024147913
	Bottom Extra	8	4	0.5		8.4675		8.4675	0.333333333	22.58	0.010245009
	Top Through	2	4	0.5		78.833	1	79.833	0.333333333	53.222	0.024147913
	Top Extra	7	4	0.5		5.648		5.648	0.333333333	13.17866667	0.005979431
	Stirrups	148	3	0.375		4.333	0.75	5.083	0.25	188.071	0.08533167
	Beam(PB-04) Between Them in Boundry Wall Left Side in long way										
	Bottom Through	3	4	0.5		9.667	1	10.667	0.333333333	10.667	0.004839837
	Bottom Extra		4	0.5							
	Top Through	2	4	0.5		9.667	1	10.667	0.333333333	7.111333333	0.003226558
	Top Extra		4	0.5							

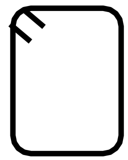
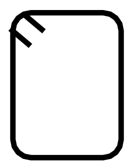
	Stirrups	19	3	0.375		5.1667	0.75	5.9167	0.25	28.104325	0.012751509
	Total									376.156325	0.170669839
	External Wall										
1.5	PB-01 6"x24" At External Wall are 11(Simply Supported)										
	Bottom Through	2	4	0.5		90.75	1	91.75	0.333333333	61.16666667	0.027752571
	Top Through	2	4	0.5		90.75	1	91.75	0.333333333	61.16666667	0.027752571
	Stirrups	171	3	0.375		4.333	0.75	5.083	0.25	217.29825	0.098592672
	Total									339.6315833	0.154097815
1.6	PB-02 6"x24" At External Wall are 2(Simply Supported)										
	Bottom Through	2	4	0.5		12.25	1	13.25	0.333333333	8.833333333	0.004007864
	Top Through	2	4	0.5		12.25	1	13.25	0.333333333	8.833333333	0.004007864
	Stirrups	25	3	0.375		4.333	0.75	5.083	0.25	31.76875	0.014414133
	Total									49.43541667	0.022429862
1.7	PB-03 6"x24" At External Wall are 1(Simply Supported)										

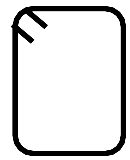
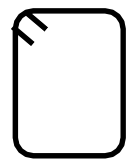
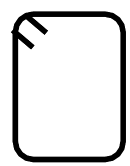
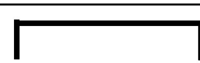
	Bottom Through	2	4	0.5		7.41667	1	8.41667	0.333333333	5.611113333	0.002545877
	Top Through	2	4	0.5		7.41667	1	8.41667	0.333333333	5.611113333	0.002545877
	Stirrups	16	3	0.375		4.333	0.75	5.083	0.25	20.332	0.009225045
	Total									31.55422667	0.0143168
1.8	PB-05 6"x24" At External Wall are 2(continuous)										
	Bottom Through	2	4	0.5		58.167	1	59.167	0.333333333	39.44466667	0.017896854
	Bottom Extra	4	4	0.5		28.993		28.993	0.333333333	38.65733333	0.017539625
	Top Through	2	4	0.5		58.167	1	59.167	0.333333333	39.44466667	0.017896854
	Stirrups	72	3	0.375		4.333	0.75	5.083	0.25	91.494	0.041512704
	Total									209.0406667	0.094846038
	Internal Wall										
1.9	PB-03 6"x24" At internal Wall are 9 (Simply supported)										
	Bottom Through	2	4	0.5		105.667	1	106.667	0.333333333	71.11133333	0.03226467
	Top Through	2	4	0.5		105.667	1	106.667	0.333333333	71.11133333	0.03226467

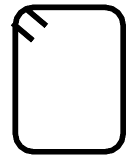
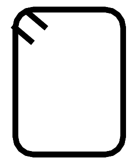
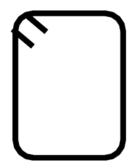
	Stirrups	193	3	0.375		4.333	0.75	5.083	0.25	245.25475	0.11127711
	Total									387.4774167	0.17580645
1.1.10	PB-02 6"x24" At internal Wall are 15 (Simply supported)										
	Bottom Through	2	4	0.5		145.13667	1	146.13667	0.333333333	97.42444667	0.044203469
	Top Through	2	4	0.5		145.13667	1	146.13667	0.333333333	97.42444667	0.044203469
	Stirrups	263	3	0.375		4.333	0.75	5.083	0.25	334.20725	0.151636683
	Total									529.0561433	0.240043622
1.1.11	PB-06 8"x24" At internal Wall are 3(Simply supported)										
	Bottom Through	2	4	0.5		4.75	1	5.75	0.333333333	3.833333333	0.001739262
	Top Through	2	4	0.5		4.75	1	5.75	0.333333333	3.833333333	0.001739262
	Stirrups	13	3	0.375		4.25	0.75	5	0.25	16.25	0.007372958
	Total									23.91666667	0.010851482
	Total Reinforcement in All Plinth Beams									5226.79381	2.371503544

BAR BENDING SCHEDULE (BBS)											
S NO	Description Of Work	Number	Bar Number	Diameter (in)	Shape	SL (ft)	HL (ft) (12Dia)	Total Length (ft)	Wt/ft	Weight (lbs)	Weight (tonnes)
1	First Floor Beam R/f										
1.1	FB-01 6"X30" GA TO GB on G1										
	Bottom Through	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Top Through	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									32.72905833	0.014849845
1.2	FB-01 6"X30" GC TO GF										
	Bottom Through	2	4	0.5		6	1	7	0.33333	4.666666667	0.002117362
	Top Through	2	4	0.5		6	1	7	0.33333	4.666666667	0.002117362
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									25.79158333	0.01170217
1.3	FB-01 6"X30" GF TO GM										
	Bottom Through	2	4	0.5		16	1	17	0.33333	11.33333333	0.005142166
	Top Through	2	4	0.5		16	1	17	0.33333	11.33333333	0.005142166
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848

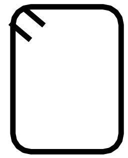
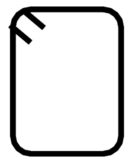
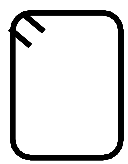
	Total									68.74976667	0.031193179
1.4	FB-01 6"X30" GN TO GP										
	Bottom Through	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Top Through	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									24.45825	0.01109721
1.5	FB-02 6"X30" GA TO GF on G2										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									66.41643333	0.030134498
1.6	FB-03 6"X15" GM to GP										
	Bottom Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Top Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Bottom Extra	2	4	0.5		5.33		5.33	0.33333	3.553333333	0.00161222
	Stirrups	16	3	0.375		3.5	0.75	4.25	0.25	17	0.007713249

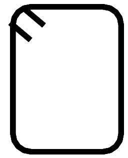
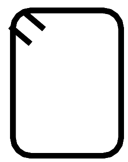
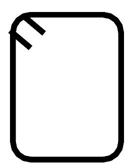
	Total									31.55333333	0.014316394
1.7	FB-04 6"X15" GF to GP										
	Bottom Through	2	4	0.5		22.25	1	23.25	0.33333	15.5	0.007032668
	Top Through	2	4	0.5		22.25	1	23.25	0.33333	15.5	0.007032668
	Top Extra Curtailed	2	4	0.5		3.75	1	4.75	0.33333	3.166666667	0.001436782
	Stirrups	43	3	0.375		5.8333	0.75	6.5833	0.25	70.770475	0.032110016
	Total									104.9371417	0.047612133
1.8	FB-06 6"X30"										
	Bottom Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Bottom Extra	2	4	0.5		4.833		4.833	0.33333	3.222	0.001461887
	Top Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									35.617725	0.016160492
1.9	FB-07 6"X30" GH to GP										
	Bottom Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Top Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026

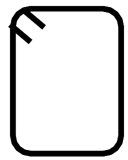
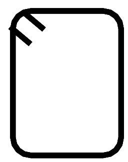
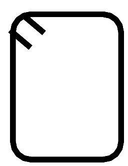
	Stirrups	31	3	0.375		5.8333	0.75	6.5833	0.25	51.020575	0.023149081
	Total									74.68724167	0.033887133
1.1.1	FB-08 6"X30" GA to GF on G8										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Bottom Extra	2	4	0.5		9		9	0.33333	6	0.002722323
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									72.41643333	0.032856821
1.1.2	FB-09 6"X30" GH to GP										
	Bottom Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Top Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Stirrups	31	3	0.375		5.8333	0.75	6.5833	0.25	51.020575	0.023149081
	Total									74.68724167	0.033887133
1.1.3	FB-10 6"X30"										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825

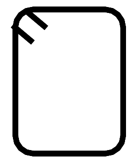
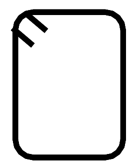
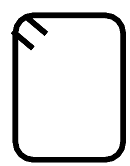
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									66.41643333	0.030134498
1.1.4	FB-12 6"X30"										
	Bottom Through	2	4	0.5		9.0833	1	10.0833	0.33333	6.7222	0.00305
	Top Through	2	4	0.5		9.0833	1	10.0833	0.33333	6.7222	0.00305
	Stirrups	19	3	0.375		5.8333	0.75	6.5833	0.25	31.270675	0.014188147
	Total									44.715075	0.020288147
1.1.5	FB-13 6"X30"										
	Bottom Through	2	4	0.5		10.6167	1	11.6167	0.33333	7.744466667	0.003513823
	Top Through	2	4	0.5		10.6167	1	11.6167	0.33333	7.744466667	0.003513823
	Stirrups	19	3	0.375		5.8333	0.75	6.5833	0.25	31.270675	0.014188147
	Total									46.75960833	0.021215793
1.1.6	FB-14 6"X30"										
	FROM GG TO GJ										
	Bottom Through	2	4	0.5		4.75	1	5.75	0.33333	3.833333333	0.001739262
	Bottom Extra	2	4	0.5		3.1667		3.1667	0.33333	2.111133333	0.000957864

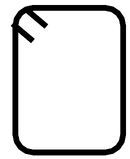
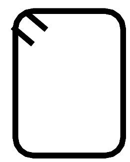
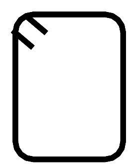
	Top Through	2	4	0.5		4.75	1	5.75	0.33333	3.833333333	0.001739262
	Top Extra curtailed	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Stirrups	10	3	0.375		4.333	0.75	5.083	0.25	12.7075	0.005765653
	Total									26.4853	0.012016924
	FROM GG TO GJ										
	Bottom Through	2	4	0.5		16.25	1	17.25	0.33333	11.5	0.005217786
	Bottom Extra	2	4	0.5		10.667		10.667	0.33333	7.111333333	0.003226558
	Top Through	2	4	0.5		16.25	1	17.25	0.33333	11.5	0.005217786
	Top Extra curtailed	2	4	0.5		4	1	5	0.33333	3.333333333	0.001512402
	Stirrups	10	3	0.375		4.333	0.75	5.083	0.25	12.7075	0.005765653
	Total									46.15216667	0.020940185
1.1.7	FB-15 6"X30"										
	FROM BOUNDRY TO GG										
	Bottom Through	2	5	0.5		22.583	1	23.583	0.41667	19.6525	0.008916742
	Bottom Extra	2	4	0.5		10.1		10.1	0.33333	6.733333333	0.003055051
	Top Through	2	5	0.5		22.583	1	23.583	0.41667	19.6525	0.008916742

	Top Extra curtailed	2	4	0.5		7.58	1	8.58	0.33333	5.72	0.002595281
	Stirrups	40	3	0.375		4.833	0.75	5.583	0.25	55.83	0.025331216
	Total									107.5883333	0.048815033
1.1.8	FB-16 6"X30"	G8 TO 9									
	Bottom Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Top Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Stirrups	22	3	0.375		5.8333	0.75	6.5833	0.25	36.20815	0.01642838
	Total									52.87481667	0.023990389
	G9 TO G11										
	Bottom Through	2	4	0.5		13.25	1	14.25	0.33333	9.5	0.004310345
	Top Through	2	4	0.5		13.25	1	14.25	0.33333	9.5	0.004310345
	Stirrups	25	3	0.375		5.8333	0.75	6.5833	0.25	41.145625	0.018668614
	Total									60.145625	0.027289304
1.1.9	FB-17 6"X30"	G4 TO G6									
	Bottom Through	2	4	0.5		5.25	1	6.25	0.33333	4.166666667	0.001890502

	Top Through	2	4	0.5		5.25	1	6.25	0.33333	4.166666667	0.001890502
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									24.79158333	0.01124845
	G6 TO G8										
	Bottom Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Top Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									31.72905833	0.014396124
1.1.10	FB-18 6"X15"	G1 TO G2									
	Bottom Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Top Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Stirrups	16	3	0.375		3.333	0.75	4.083	0.25	16.332	0.007410163
	Total									26.66533333	0.012098609
1.1.11	FB-19 6"X30"	G2 TO G4									

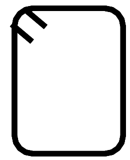
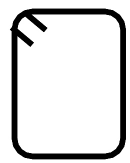
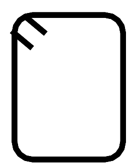
	Bottom Through	2	4	0.5		15.75	1	16.75	0.33333	11.16666667	0.005066546
	Top Extra curtailed	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Top Through	2	4	0.5		15.75	1	16.75	0.33333	11.16666667	0.005066546
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									74.081	0.033612069
1.1.12	FB-20 6"X18"										
	Bottom Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Top Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Stirrups	25	3	0.375		3.833	0.75	4.583	0.25	28.64375	0.012996257
	Total									45.31041667	0.020558265
1.1.13	FB-21 6"X30"										
	Bottom Through	2	4	0.5		14.75	1	15.75	0.33333	10.5	0.004764065
	Top Through	2	4	0.5		14.75	1	15.75	0.33333	10.5	0.004764065
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									67.081	0.030436025

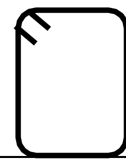
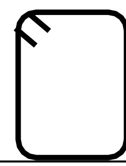
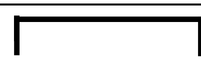
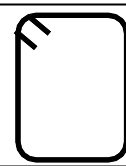
1.1.14	FB-22 6"X30"	FROM G3 TO G6									
	Bottom Through	2	4	0.5		13.75	1	14.75	0.33333	9.833333333	0.004461585
	Top Through	2	4	0.5		13.75	1	14.75	0.33333	9.833333333	0.004461585
	Stirrups	25	3	0.375		5.833	0.75	6.583	0.25	41.14375	0.018667763
	Total									60.81041667	0.027590933
	FROM G6 TO G8										
	Bottom Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Top Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Stirrups	13	3	0.375		5.833	0.75	6.583	0.25	21.39475	0.009707237
	Total									31.06141667	0.014093202
1.1.15	FB-23 6"X30"	FROM G9 TO G12									
	Bottom Through	2	4	0.5		15.25	1	16.25	0.33333	10.83333333	0.004915306
	Top Through	2	4	0.5		15.25	1	16.25	0.33333	10.83333333	0.004915306
	Stirrups	25	3	0.375		5.8333	0.75	6.5833	0.25	41.145625	0.018668614
	Total									62.81229167	0.028499225
1.1.15	FB-24 6"X30"	FROM G10 TO G12									

	Bottom Through	2	4	0.5		9.583	1	10.583	0.33333	7.055333333	0.003201149
	Top Through	2	4	0.5		9.583	1	10.583	0.33333	7.055333333	0.003201149
	Stirrups	16	3	0.375		5.8333	0.75	6.5833	0.25	26.3332	0.011947913
	Total									40.44386667	0.018350212
	FROM G12 TO G13										
	Bottom Through	2	4	0.5		7.833	1	8.833	0.33333	5.888666667	0.002671809
	Top Through	2	4	0.5		7.833	1	8.833	0.33333	5.888666667	0.002671809
	Stirrups	12	3	0.375		5.8333	0.75	6.5833	0.25	19.7499	0.008960935
	Total									31.52723333	0.014304552
1.1.16	FB-25 6"X30"										
	FROM G6 TO G8										
	Bottom Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Top Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Stirrups	13	3	0.375		5.833	0.75	6.583	0.25	21.39475	0.009707237
	Total									31.06141667	0.014093202

1.1.17	FB-26 6"X30"										
	Bottom Through	2	4	0.5		29.5	1	30.5	0.33333	20.33333333	0.00922565
	Top Through	2	4	0.5		29.5	1	30.5	0.33333	20.33333333	0.00922565
	Stirrups	54	3	0.375		5.833	0.75	6.583	0.25	88.8705	0.040322368
	Total									129.5371667	0.058773669
1.1.18	FB-27 6"X30"										
	Bottom Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Top Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Stirrups	16	3	0.375		5.833	0.75	6.583	0.25	26.332	0.011947368
	Total									39.99866667	0.018148215
1.1.19	FB-28 6"X30"										
	FROM G8 TO G10										
	Bottom Through	2	5	0.5		17.667	1	18.667	0.41667	15.55583333	0.007058001
	Bottom Extra	2	4	0.5		11.77		11.77	0.33333	7.846666667	0.003560194
	Top Through	2	5	0.5		17.667	1	18.667	0.41667	15.55583333	0.007058001
	Top Extra curtailed	2	4	0.5		8.833	1	9.833	0.33333	6.555333333	0.002974289

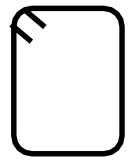
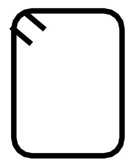
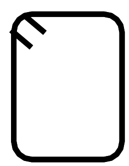
	Stirrups	31	3	0.375		5.833	0.75	6.583	0.25	51.01825	0.023148026
	Total									96.53191667	0.04379851
	FROM G10 TO G13										
	Bottom Through	2	5	0.5		15.0833	1	16.0833	0.41667	13.40275	0.006081103
	Bottom Extra	2	4	0.5		10		10	0.33333	6.666666667	0.003024803
	Top Through	2	5	0.5		15.0833	1	16.0833	0.41667	13.40275	0.006081103
	Top Extra curtailed	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									85.21983333	0.038665986
1.1.20	FB-29 8"X24"										
	Bottom Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Top Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Stirrups	16	3	0.375		5.1667	0.75	5.9167	0.25	23.6668	0.010738113
	Total									37.33346667	0.016938959
1.1.21	FB-1a 4"X30"										

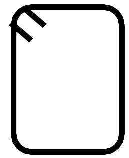
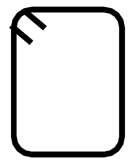
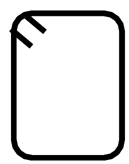
	Bottom Through	2	4	0.5		7	1	8	0.33333	5.333333333	0.002419843
	Top Through	2	4	0.5		7	1	8	0.33333	5.333333333	0.002419843
	Stirrups	16	3	0.375		5.5	0.75	6.25	0.25	25	0.011343013
	Total									35.66666667	0.016182698
1.1.22	FB-11a 6"X30"										
	Bottom Through	2	4	0.5		8.75	1	9.75	0.33333	6.5	0.002949183
	Top Through	2	4	0.5		8.75	1	9.75	0.33333	6.5	0.002949183
	Stirrups	16	3	0.375		5.833	0.75	6.583	0.25	26.332	0.011947368
	Total									39.332	0.017845735
1.1.23	CB-11a 18"X6"										
	Bottom Through	4	4	0.5		5.25	1	6.25	0.33333	8.333333333	0.003781004
	Top Through	4	4	0.5		5.25	1	6.25	0.33333	8.333333333	0.003781004
	Stirrups	7	3	0.375		3.833	0.75	4.583	0.25	8.02025	0.003638952
	Total									24.68691667	0.01120096
											1.886445765

BAR BENDING SCHEDULE (BBS)											
S NO	Description Of Work	Number	Bar Number	Diameter (in)	Shape	SL (ft)	HL (ft) (12Dia)	Total Length (ft)	Wt/ft	Weight (lbs)	Weight (tonnes)
1	First Floor Beam R/f										
1.1	RB-01 6"X30" GA TO GB on G1										
	Bottom Through	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Top Through	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									32.72905833	0.014849845
1.2	RB-01 6"X30" GC TO GF										
	Bottom Through	2	4	0.5		6	1	7	0.33333	4.666666667	0.002117362
	Top Through	2	4	0.5		6	1	7	0.33333	4.666666667	0.002117362
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									25.79158333	0.01170217
1.3	RB-01 6"X30" GF TO GM										
	Bottom Through	2	4	0.5		16	1	17	0.33333	11.33333333	0.005142166
	Top Through	2	4	0.5		16	1	17	0.33333	11.33333333	0.005142166
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848

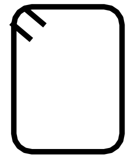
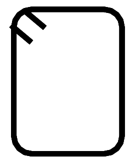
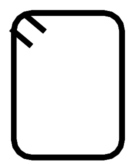
	Total									68.74976667	0.031193179
1.4	RB-01 6"X30" GN TO GP										
	Bottom Through	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Top Through	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									24.45825	0.01109721
1.5	RB-02 6"X30" GA TO GF on G2										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									66.41643333	0.030134498
1.6	RB-03 6"X15" GM to GP										
	Bottom Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Top Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Bottom Extra	2	4	0.5		5.33		5.33	0.33333	3.553333333	0.00161222
	Stirrups	16	3	0.375		3.5	0.75	4.25	0.25	17	0.007713249

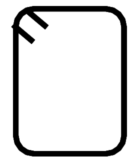
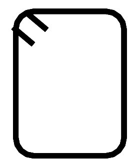
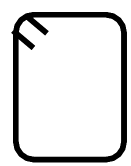
	Total									31.55333333	0.014316394
1.7	RB-04 6"X15" GF to GP										
	Bottom Through	2	4	0.5		22.25	1	23.25	0.33333	15.5	0.007032668
	Top Through	2	4	0.5		22.25	1	23.25	0.33333	15.5	0.007032668
	Top Extra Curtailed	2	4	0.5		3.75	1	4.75	0.33333	3.166666667	0.001436782
	Stirrups	43	3	0.375		5.8333	0.75	6.5833	0.25	70.770475	0.032110016
	Total									104.9371417	0.047612133
1.8	RB-06 6"X30"										
	Bottom Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Bottom Extra	2	4	0.5		4.833		4.833	0.33333	3.222	0.001461887
	Top Through	2	4	0.5		7.25	1	8.25	0.33333	5.5	0.002495463
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									35.617725	0.016160492
1.9	RB-07 6"X30" GH to GP										
	Bottom Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Top Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026

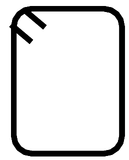
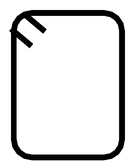
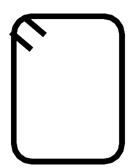
	Stirrups	31	3	0.375		5.8333	0.75	6.5833	0.25	51.020575	0.023149081
	Total									74.68724167	0.033887133
1.1.1	RB-08 6"X30" GA to GF on G8										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Bottom Extra	2	4	0.5		9		9	0.33333	6	0.002722323
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									72.41643333	0.032856821
1.1.2	RB-09 6"X30" GH to GP										
	Bottom Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Top Through	2	4	0.5		16.75	1	17.75	0.33333	11.83333333	0.005369026
	Stirrups	31	3	0.375		5.8333	0.75	6.5833	0.25	51.020575	0.023149081
	Total									74.68724167	0.033887133
1.1.3	RB-10 6"X30"										
	Bottom Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825
	Top Through	2	4	0.5		14.25	1	15.25	0.33333	10.16666667	0.004612825

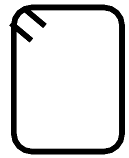
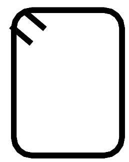
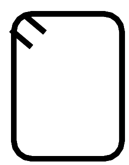
	Stirrups	28	3	0.375		5.8333	0.75	6.5833	0.25	46.0831	0.020908848
	Total									66.41643333	0.030134498
1.1.4	RB-12 6"X30"										
	Bottom Through	2	4	0.5		9.0833	1	10.0833	0.33333	6.7222	0.00305
	Top Through	2	4	0.5		9.0833	1	10.0833	0.33333	6.7222	0.00305
	Stirrups	19	3	0.375		5.8333	0.75	6.5833	0.25	31.270675	0.014188147
	Total									44.715075	0.020288147
1.1.5	RB-13 6"X30"										
	Bottom Through	2	4	0.5		10.6167	1	11.6167	0.33333	7.744466667	0.003513823
	Top Through	2	4	0.5		10.6167	1	11.6167	0.33333	7.744466667	0.003513823
	Stirrups	19	3	0.375		5.8333	0.75	6.5833	0.25	31.270675	0.014188147
	Total									46.75960833	0.021215793
1.1.6	RB-14 6"X30"										
	FROM GG TO GJ										
	Bottom Through	2	4	0.5		4.75	1	5.75	0.33333	3.833333333	0.001739262
	Bottom Extra	2	4	0.5		3.1667		3.1667	0.33333	2.111133333	0.000957864

	Top Through	2	4	0.5		4.75	1	5.75	0.33333	3.833333333	0.001739262
	Top Extra curtailed	2	4	0.5		5	1	6	0.33333	4	0.001814882
	Stirrups	10	3	0.375		4.333	0.75	5.083	0.25	12.7075	0.005765653
	Total									26.4853	0.012016924
	FROM GG TO GJ										
	Bottom Through	2	4	0.5		16.25	1	17.25	0.33333	11.5	0.005217786
	Bottom Extra	2	4	0.5		10.667		10.667	0.33333	7.111333333	0.003226558
	Top Through	2	4	0.5		16.25	1	17.25	0.33333	11.5	0.005217786
	Top Extra curtailed	2	4	0.5		4	1	5	0.33333	3.333333333	0.001512402
	Stirrups	10	3	0.375		4.333	0.75	5.083	0.25	12.7075	0.005765653
	Total									46.15216667	0.020940185
1.1.7	RB-15 6"X30"										
	FROM BOUNDRY TO GG										
	Bottom Through	2	5	0.5		22.583	1	23.583	0.41667	19.6525	0.008916742
	Bottom Extra	2	4	0.5		10.1		10.1	0.33333	6.733333333	0.003055051
	Top Through	2	5	0.5		22.583	1	23.583	0.41667	19.6525	0.008916742

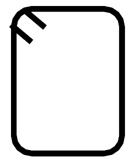
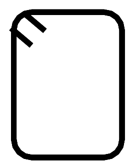
	Top Extra curtailed	2	4	0.5		7.58	1	8.58	0.33333	5.72	0.002595281
	Stirrups	40	3	0.375		4.833	0.75	5.583	0.25	55.83	0.025331216
	Total									107.5883333	0.048815033
1.1.8	RB-16 6"X30"	G8 TO 9									
	Bottom Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Top Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Stirrups	22	3	0.375		5.8333	0.75	6.5833	0.25	36.20815	0.01642838
	Total									52.87481667	0.023990389
	G9 TO G11										
	Bottom Through	2	4	0.5		13.25	1	14.25	0.33333	9.5	0.004310345
	Top Through	2	4	0.5		13.25	1	14.25	0.33333	9.5	0.004310345
	Stirrups	25	3	0.375		5.8333	0.75	6.5833	0.25	41.145625	0.018668614
	Total									60.145625	0.027289304
1.1.9	RB-17 6"X30"	G4 TO G6									
	Bottom Through	2	4	0.5		5.25	1	6.25	0.33333	4.166666667	0.001890502

	Top Through	2	4	0.5		5.25	1	6.25	0.33333	4.166666667	0.001890502
	Stirrups	10	3	0.375		5.8333	0.75	6.5833	0.25	16.45825	0.007467446
	Total									24.79158333	0.01124845
	G6 TO G8										
	Bottom Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Top Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Stirrups	13	3	0.375		5.8333	0.75	6.5833	0.25	21.395725	0.009707679
	Total									31.72905833	0.014396124
1.1.10	RB-18 6"X15"	G1 TO G2									
	Bottom Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Top Through	2	4	0.5		6.75	1	7.75	0.33333	5.166666667	0.002344223
	Stirrups	16	3	0.375		3.333	0.75	4.083	0.25	16.332	0.007410163
	Total									26.66533333	0.012098609
1.1.11	RB-19 6"X30"	G2 TO G4									

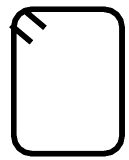
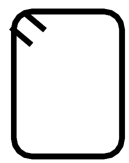
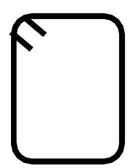
	Bottom Through	2	4	0.5		15.75	1	16.75	0.33333	11.16666667	0.005066546
	Top Extra curtailed	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Top Through	2	4	0.5		15.75	1	16.75	0.33333	11.16666667	0.005066546
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									74.081	0.033612069
1.1.12	RB-20 6"X18"										
	Bottom Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Top Through	2	4	0.5		11.5	1	12.5	0.33333	8.333333333	0.003781004
	Stirrups	25	3	0.375		3.833	0.75	4.583	0.25	28.64375	0.012996257
	Total									45.31041667	0.020558265
1.1.13	RB-21 6"X30"										
	Bottom Through	2	4	0.5		14.75	1	15.75	0.33333	10.5	0.004764065
	Top Through	2	4	0.5		14.75	1	15.75	0.33333	10.5	0.004764065
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									67.081	0.030436025

1.1.14	RB-22 6"X30"	FROM G3 TO G6									
	Bottom Through	2	4	0.5		13.75	1	14.75	0.33333	9.833333333	0.004461585
	Top Through	2	4	0.5		13.75	1	14.75	0.33333	9.833333333	0.004461585
	Stirrups	25	3	0.375		5.833	0.75	6.583	0.25	41.14375	0.018667763
	Total									60.81041667	0.027590933
	FROM G6 TO G8										
	Bottom Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Top Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Stirrups	13	3	0.375		5.833	0.75	6.583	0.25	21.39475	0.009707237
	Total									31.06141667	0.014093202
1.1.15	RB-23 6"X30"	FROM G9 TO G12									
	Bottom Through	2	4	0.5		15.25	1	16.25	0.33333	10.83333333	0.004915306
	Top Through	2	4	0.5		15.25	1	16.25	0.33333	10.83333333	0.004915306
	Stirrups	25	3	0.375		5.8333	0.75	6.5833	0.25	41.145625	0.018668614
	Total									62.81229167	0.028499225
1.1.15	RB-24 6"X30"	FROM G10 TO G12									

	Bottom Through	2	4	0.5		9.583	1	10.583	0.33333	7.055333333	0.003201149
	Top Through	2	4	0.5		9.583	1	10.583	0.33333	7.055333333	0.003201149
	Stirrups	16	3	0.375		5.8333	0.75	6.5833	0.25	26.3332	0.011947913
	Total									40.44386667	0.018350212
	FROM G12 TO G13										
	Bottom Through	2	4	0.5		7.833	1	8.833	0.33333	5.888666667	0.002671809
	Top Through	2	4	0.5		7.833	1	8.833	0.33333	5.888666667	0.002671809
	Stirrups	12	3	0.375		5.8333	0.75	6.5833	0.25	19.7499	0.008960935
	Total									31.52723333	0.014304552
1.1.16	RB-25 6"X30"										
	FROM G6 TO G8										
	Bottom Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Top Through	2	4	0.5		6.25	1	7.25	0.33333	4.833333333	0.002192982
	Stirrups	13	3	0.375		5.833	0.75	6.583	0.25	21.39475	0.009707237
	Total									31.06141667	0.014093202

1.1.17	RB-26 6"X30"										
	Bottom Through	2	4	0.5		29.5	1	30.5	0.33333	20.33333333	0.00922565
	Top Through	2	4	0.5		29.5	1	30.5	0.33333	20.33333333	0.00922565
	Stirrups	54	3	0.375		5.833	0.75	6.583	0.25	88.8705	0.040322368
	Total									129.5371667	0.058773669
1.1.18	RB-27 6"X30"										
	Bottom Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Top Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Stirrups	16	3	0.375		5.833	0.75	6.583	0.25	26.332	0.011947368
	Total									39.99866667	0.018148215
1.1.19	RB-28 6"X30"										
	FROM G8 TO G10										
	Bottom Through	2	5	0.5		17.667	1	18.667	0.41667	15.55583333	0.007058001
	Bottom Extra	2	4	0.5		11.77		11.77	0.33333	7.846666667	0.003560194
	Top Through	2	5	0.5		17.667	1	18.667	0.41667	15.55583333	0.007058001
	Top Extra curtailed	2	4	0.5		8.833	1	9.833	0.33333	6.555333333	0.002974289

	Stirrups	31	3	0.375		5.833	0.75	6.583	0.25	51.01825	0.023148026
	Total									96.53191667	0.04379851
	FROM G10 TO G13										
	Bottom Through	2	5	0.5		15.0833	1	16.0833	0.41667	13.40275	0.006081103
	Bottom Extra	2	4	0.5		10		10	0.33333	6.666666667	0.003024803
	Top Through	2	5	0.5		15.0833	1	16.0833	0.41667	13.40275	0.006081103
	Top Extra curtailed	2	4	0.5		7.5	1	8.5	0.33333	5.666666667	0.002571083
	Stirrups	28	3	0.375		5.833	0.75	6.583	0.25	46.081	0.020907895
	Total									85.21983333	0.038665986
1.1.20	RB-29 8"X24"										
	Bottom Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Top Through	2	4	0.5		9.25	1	10.25	0.33333	6.833333333	0.003100423
	Stirrups	16	3	0.375		5.1667	0.75	5.9167	0.25	23.6668	0.010738113
	Total									37.33346667	0.016938959
1.1.21	RB-1a 4"X30"										

	Bottom Through	2	4	0.5		7	1	8	0.33333	5.333333333	0.002419843
	Top Through	2	4	0.5		7	1	8	0.33333	5.333333333	0.002419843
	Stirrups	16	3	0.375		5.5	0.75	6.25	0.25	25	0.011343013
	Total									35.66666667	0.016182698
1.1.22	RB-11a 6"X30"										
	Bottom Through	2	4	0.5		8.75	1	9.75	0.33333	6.5	0.002949183
	Top Through	2	4	0.5		8.75	1	9.75	0.33333	6.5	0.002949183
	Stirrups	16	3	0.375		5.833	0.75	6.583	0.25	26.332	0.011947368
	Total									39.332	0.017845735
1.1.23	RB-11a 18"X6"										
	Bottom Through	4	4	0.5		5.25	1	6.25	0.33333	8.333333333	0.003781004
	Top Through	4	4	0.5		5.25	1	6.25	0.33333	8.333333333	0.003781004
	Stirrups	7	3	0.375		3.833	0.75	4.583	0.25	8.02025	0.003638952
	Total									24.68691667	0.01120096
											1.391884407

RATE ANALYSIS FOR SITE DEVELOPNMENT

Site Clearance

Labors(Unskilled)	4	
Rate	1100	per day
Cost	4400	Rs.

Equipment Cost			
Machinery	25000	Rs	
Manual Tools	3000	Rs	
Fuel + Others	7000	Rs	
Total equipment cost	35000	Rs	
Transportation to dump material away from site =		9000	Rs.

Survey

Skilled Surveyor	2200	Rs./day
Helper	1400	Rs./day
Total Cost =	52000	
Profit Magins 10%	5200	Rs
Total Price For Site Developnment =	**57200**	**Rs.**

Plot Area =	4500	sq.ft
Price =	12.71111	Rs/sft
Price =	57200	Rs.

RATE ANALYSIS FOR EXCAVATION FOR FOUNDATION AND UGWT (MANUAL)

Crew Size

Skilled Labor	1(Supervison/Mistri	
Unskilled Labor	5(Laborers)	
Productivity Assumptions		
1 Laborer =	200	cft/day
5 Laborer =	1000	cft/day

Labor Cost Calculation

Labor	Quantity	Rate(Rs/day)	Cost
Skilled	1	1300	1300
Unskilled	5	1100	5500
		Total	Rs.6800

Output/Day = 1,000 cft

Labor Cost per cft = Rs. 6,800 / 1,000 = Rs.6.80/cft

Equipment Cost

Shovels, buckets, wheelbarrows, etc		
Lump Sum =	0.5/cft	

Overheads and Profits

Overheads =	5%	
Profit =	10%	
Total	Rs.1.095	

Item	**Rate(Rs./cft)**	
Labor	6.8	
Equipment	0.5	
Overheads+profit	1.095	
Total Rate per cft	8.395	Rs./cft

Excavation for footing and UGWT = 7464.28 cft

Price For Excavation for Footing and UGWT = 62662.63 Rs.

RATE ANALYSIS FOR IMPORTED EARTH FILL

Backfill of Soil = 3663.54 cft

Taking 20% of backfill Soil to buy while rest will be covered using excavated soil

Item	Rate(Rs./cft)
Borrow earth(Excavated)	9
Loading	2.5
Transportation	7
Spreading,water,compaction	5
Total Rate/cft	**23.5**
Total Material Cost =	**17218.638**

Labor Cost

Unskilled labor	4	1200	Rs/day
Total Labor Cost =	**4800**	**Rs.**	

Equipment Cost

Compactor	2000	per day
Tools	1000	per day
Total Equipment Cost =	**3000**	**Rs./ day**

Total Cost of imported earth fill = 25018.638 Rs

Profit Margins 10% 2501.8638 Rs

Total Price of imported earth fill = 27520.5018 Rs

RATE ANALYSIS FOR TERMITE PROOFING

Material

Item	Quantity per sft	Rate(Rs.)	Cost/sft
Termicitide	0.1 liters	500/liters	50

Labor

Type	Rate/day	Productivity sft/day	Cost/sft	
Skilled sprayer	1800	1500	1.5	
Helper	1200		0.5	
		Total Labor Cost =	2	Rs./ sft

Equipment

Item	Rate	Cost	
Tools	Lump Sum	1	Rs/sft

Over Heads + Profit (15%) 4.8 Rs/sft

Total Rate =	57.8	Rs./sft

For Quantity

65% of plot area = 2925 sft

Total Price for Termite Proofing = 169065 Rs

RATE ANALYSIS FOR LEAN LAYING(1:4:8)

Material Cost

CEMENT BAGS:

Volume =	1115.29	cft(Wet Volume)
Volume =	1728.6995	cft(Dry Volume)
Cement =	132.9768846	cft
Volume of 1 bag =	1.25	cft
No. of bags =	106.3815077	
No. of bags =	107	Bags

AGGREGATES:

Fine Aggregates =	531.9075385	cft
Coarse Aggregates =	1063.815077	cft
Water		
Volume Of Water =	66.48844231	cft
1 cft =	7.418	US gallons
Volume Of Water =	493.211265	Gallons

ACTIVITY DURATION AND LABOR REQUIREMENT

Labor Productivity =	0.8	lbhr/cyd
Productivity = Labours hour/unit		
Labor hours =	33.04562963	hrs
10 hours a day		
Duration =	3.304562963	days

ITEM	QUANTITY	RATE	COST	
1-Material				
Cement	107 bags	Rs.1300/bag	139100	
Fine Aggregate	531.91 cft	Rs.30/cft	15957.3	
Coarse Aggregate	1063.81 cft	Rs 55/cft	58509.55	
Water	493.211 gallons	Rs.1560/1000 gallons	769.4092	
		Total material cost =	214336.3	
2-Labor Cost				
Skilled(mason)	1	Rs.2083/day	6883.405	
Unskilled(helper)	3	Rs.1400/day	13879.16	
		Total labor cost =	20762.57	
		Total Cost =	235098.8	Rs.
Overheads		0.05	11754.94	
Profit		0.1	23509.88	
Total price for 1115.29 cft		**Rs.**	**270364**	
Rate per cft			242.4156	per cft

RATE ANALYSIS FOR BLOCK MASONRY

Material Cost per Block

Mortar Mix =	1:06	(cement:sand)		
1 block volume =	0.375	cft (6"x9"x12")		
Mortar volume per block (with 10 mm joints & waste) =			0.15	cft/Block

Mortal Dry Volume = 0.1995 cft/block

Cement

Cement Bags =	0.0285	cft/block		
Volume of 1 bag =	1.25	cft		
Cement bags =	0.0228	bags/block		
Sand =	0.171	cft/block		
Material	Qty/Block	Rate	Cost/Block Rs.	
Block	1	Rs.45/Block	45	
Cement	0.0228	1300	29.64	
Sand	0.171	70	11.97	
Total Material Cost Per Block=			**86.61**	**Rs.**

Labor Cost Per Block

Skilled Labor(Mason)	1	1800	Rs./day
Unskilled Labor(Helper)	1	1200	Rs/day
Labor Cost =		**3000**	**Rs/day**
Mason Lays =	400	Blocks/day	
Labor Cost =	**7.5**	**Rs/Block**	

Equipment Cost

Tools,Water,Curing,Scaffolding Lump Sum 2 Rs/Block

Total Resource Cost = 96.11 Rs/Block

Total Resource Cost = 96.11 Rs/Block

No. of Blocks = 7840 Blocks

Total Cost For Block Masonry = 753502.4 Rs.

Overheads (5%) 37675.12 Rs

Profit Margin (10%) 75350.24 Rs

Total Price For Block Masonry = 866527.76 Rs.

RATE ANALYSIS FOR PLASTER

Material Required

No. of Cement Bags = 231 Bags

Sand

Plaster Mix = 1:06 (cement:sand)

For every 1 bag of cement, sand needed = 7.5 cft

For 231 Bags = 1732.5 cft of sand

Material Cost

Material	Quantity	Rate	Cost	
Cement	231 bags	Rs.1200/Bag	277200	Rs.
Sand	1732.5 cft	Rs.70/cft	121275	Rs.
Total Material Cost =			**398475**	**Rs.**

Labor Cost

Skilled(Mason)	2	2000	Rs./day
Unskilled (Helper)	2	1400	Rs./day
Labor Cost =		6800	Rs/day
1 mason + 1 helper plaster = 100 SFT/day			
2 mason + 2 helper plaster = 200 SFT/day			
Total Plaster =	26908.91	sft	
Workdays =	134.54455		
Workdays =	135	Days	
Total Labor Cost =	**918000**	**Rs.**	

Equipment Cost

Tools, Curing, Water, Scaffolding

Lump Sum = 10000 Rs.

Total Cost Of Plaster =	**1326475**	**Rs.**
Overheads 5%	66323.75	Rs
Profit 10%	132647.5	Rs
Total Price Of Plaster =	**1525446.3**	**Rs**

RATE ANALYSIS FOR MARBLE FLOORING

Item	Area(sft)
Flooring	4307.44
Skirting	392.61
Steps	243.64
Total	**4943.69**

Material Cost

Material	Rate(Rs.)	Cost/sft	
Marble(Best Quality Local)	90	90	
Cement	1300/bag	6	
Sand	70/cft	3	
White Cement and Polish		2	
Total Material Cost =		**101**	**Rs./sft**

Labor Cost

Marble layer + helper =	3000	Rs/day				
Polisher + helper	3000	Rs./day				
Productivity =	150	sft/day/crew(laying+polishing)				
Labor Cost =	**40**	**Rs/sft**	**(Rs 30 for Laying+Rs 30 for Polishing)**			

Equipment Cost

Tools, water, etc. =	**5**	**Rs/sft**
Total Resource Cost =	**146**	**Rs/sft**
Total Cost For Marble Flooring =	**721778.74**	**Rs**
Overheads =	36088.937	Rs
Profit =	72177.874	Rs
Total Price For Marble Flooring =	**830045.551**	**Rs.**

RATE ANALYSIS FOR TILE FLOORING(BATHROOM TILES 8"X8")

Total Tile Area = 1292.5 sft

Material Cost

Material	Estimation Basis	Rate(Rs.)	Cost/sft Rs.	
Ceramic Tiles	Market Rate	130/sft	130	
Tiles Adhesive	1.5kg/sft	60/kg	25	
Cement(bedding)	0.1bag/sft	1300/bag	6	
Sand(bedding)	0.5cft/SFT	70	3	
White Cement	0.02bag/sft	1300/bag	2	
Total Material Cost =			**166**	**Rs/sft**

Labor Cost

Labor Type	Productivity	Rate/day	Cost/sft	
Tile fixer+helper	150 sft/day	3000	20	
Total Labor Cost =			**20**	**Rs/sft**

Equipment Cost

Tools, water, curing, cleaning	**5**	**Rs/sft**
Total Resource Cost =	**191**	**Rs/sft**
Total Cost For Tile Flooring =	**246867.5**	**Rs.**
Overheads 5% =	12343.375	Rs.
Profit 10% =	24686.75	Rs.
Total Price For Tile Flooring =	**283897.63**	**Rs.**

RATE ANALYSIS FOR DOORS,WINDOWS AND VENTILATORS

Quantities	Area (rft)
Wooden Frames	960
Door Panels	1329.4
Bedding	2094

For Wooden Frames

Material Cost

Item	Unit	Qty/rft	Rate(Rs.)	Cost/rft	
Hardwood (Kail or Partal)	rft	1	130	130	
Nails, glue, screws	L.S	1		10	
Total Material Cost =				**140**	**Rs/rft**

Labor Cost

Labor(Carpenter+Helper)	600	Rs/day	40	rft/day	
Total Labor Cost =				**15**	**Rs/rft**

Total Frame Rate =	155	Rs/rft
Total Cost for wooden frames =	**148800**	**Rs**

For Doors Panels

Covert rft to sft

Standard Door Height =	7	ft
Door width per panel =	rft/7	
Area in sft =	189.9143	sft

Item	Unit	Qty/sft	Rate (Rs.)	Cost/sft	
Flush/panel door	sft	1	170	170	
Hinges,lock	L.S	1		15	
Polish/Paint	sft	1		25	
Fixing Labor	sft	1		30	
Total Rate per sft =				240	Rs/sft
Total Cost for Door Panels =				**45579.4**	**Rs.**

For Beddings

Usually done with cement-sand mortar 1:4 under the frame during installation.

Item	Unit	Qty/rft	Rate(Rs.)	Cost/Rft	
Cement Mortar (1:4)	rft	1	30	30	
Labor(fixing)	rft	1		10	
Total rate For Bedding =				40	Rs/rft
Total Cost For Bedding =				**83760**	**Rs.**

Total Cost For Doors,Windows & Ventilators =	**278139**	**Rs.**
Overheads 5%	13906.97	Rs.
Profit 10%	27813.94	Rs.

Total Price for Doors,Windows & Ventilators =	**319860 Rs.**

RATE ANALYSIS FOR REINFORCEMENT IN SUB STRUCTURE

# 3 bar =	1.468	tons
# 4 bar =	3.492	tons

Item	Explanation	Rate(Rs.)	
Steel	From PBS (May 2025)	240000	per ton
Cutting & Bending	Labor Charges	5000	per ton
Fixing/Placing Labor	Skilled Labor for Fixing	6000	per ton
Binding Wire	10 kg @ Rs. 100/kg	1000	per ton
Wastage 2%	2% of 240000	4800	
Transportation	Truck movement	2000	per ton
Total Rate per ton =		**258800**	**Rs/ton**

Total Cost For # 3 bar =		**379918.4**	**Rs.**
Total Cost For # 4 bar =		**903729.6**	**Rs.**
Total Cost For Steel =		**1283648**	**Rs.**
Overheads 5%		64182.4	Rs.
Profit 10%		128364.8	Rs.
Total Price for R/f for Sub Structure =		**1476195.2**	**Rs.**

RATE ANALYSIS FOR REINFORCEMENT IN SUPER STRUCTURE

# 3 bar =	4.458	tons
# 4 bar =	2.956	tons
# 5 bar =	0.044	tons

Item	Explanation	Rate(Rs.)	
Steel	From PBS (May 2025)	240000	per ton
Cutting & Bending	Labor Charges	5000	per ton
Fixing/Placing Labor	Skilled Labor for Fixing	6000	per ton
Binding Wire	10 kg @ Rs. 100/kg	1000	per ton
Wastage 2%	2% of 240000	4800	
Transportation	Truck movement	2000	per ton
Total Rate per ton =		**258800**	**Rs/ton**
Total Cost For # 3 bar =		**1153730.4**	**Rs.**
Total Cost For # 4 bar =		**765012.8**	**Rs.**
Total Cost For # 5 bar =		**11387.2**	**Rs.**
Total Cost For Steel =		**1930130.4**	**Rs.**
Overheads 5%		96506.52	Rs.
Profit 10%		193013.04	Rs.
Total Price for R/f for Super Structure =		**3329474.9**	**Rs.**

RATE ANALYSIS FOR 1:2:4 CONCRETE(SUB-STUCTURE)

DATA:

MATERIAL	**COST**	
Cement	Rs.1300/bag	
Coarse aggregate	Rs.55/cft	
Fine aggregate	Rs.30/cft	
Water Tanker	Rs.1560/1000 gallons	
w/c ratio	0.5	
Labor Productivity	0.8 lhr/cyd	
Skilled Labor	Rs.2000/day	
Unskilled Labor	Rs.1000/day	
Shuttering	Rs.60/sft	
Crew	2 Skilled Labors,4 Unskilled Labors	
Tools	Rs.1500/day	
Overheads	5%	
Profit Margin	10%	

CEMENT BAGS:

Volume =	1465.57	cft(Wet Volume)
Volume =	2271.6335	cft(Dry Volume)
Cement =	324.5190714	cft
Volume of 1 bag =	1.25	cft
No. of bags =	259.6152571	
No. of bags =	260	Bags

AGGREGATES:

Fine Aggregates =	649.0381429	cft
Coarse Aggregates =	1298.076286	cft
Water		
Volume Of Water =	162.2595357	cft
1 cft =	7.418	US gallons
Volume Of Water =	1203.641236	Gallons

FORMWORK/SHUTTERING AREA

Shuttering(sft) =	732.785	sft
Shuttering rate per sft =	60	Rs/sft

ACTIVITY DURATION AND LABOR REQUIREMENT

Productivity = Labours hour/unit		
Labor hours =	43.4242963	hrs
8 hours a day		
Duration =	5.428037037	days

ITEM	QUANTITY	RATE	COST	
1-Material				
Cement	260 bags	Rs.1300/bag	338000	Rs.
Fine Aggregate	649.038 cft	Rs.30/cft	19471.14	Rs.
Coarse Aggregate	1298.076 cft	Rs 55/cft	71394.18	Rs.
Water	1203.64 gallons	Rs.1560/1000 gallons	1877.678	Rs.
		Total Material Cost =	**430743**	**Rs.**
2-Labor Cost				
Skilled	2	Rs.2000/day	13570	Rs.
Unskilled	4	Rs.1000/day	13570	Rs.
		Total Labor Cost =	**27140**	**Rs.**
3-Equipment Cost				
Shuttering	732.785 sft	Rs.60/sft	43967.1	Rs.
Tools	Lump Sum	1500/day	5088.75	Rs.
		Total Euipment Cost=	**49055.9**	**Rs.**
		Total Resource Cost =	**506939**	**Rs.**
Overheads		5%	25346.94	Rs.
Profit		10%	50693.88	Rs.
Total price for 1465.57 cft		**Rs.**	**582980**	**Rs.**
Rate per cft			397.7836	

RATE ANALYSIS FOR 1:2:4 CONCRETE(SUPER-STUCTURE)

CEMENT BAGS:

Volume =	6057.864	cft(Wet Volume)
Volume =	9389.6892	cft(Dry Volume)
Cement =	1341.384171	cft
Volume of 1 bag =	1.25	cft
No. of bags =	1073.107337	
No. of bags =	1074	Bags

AGGREGATES:

Fine Aggregates =	2682.768343	cft
Coarse Aggregates =	5365.536686	cft
Water		
Volume Of Water =	670.6920857	cft
1 cft =	7.418	US gallons
Volume Of Water =	4975.193892	Gallons

FORMWORK/SHUTTERING AREA

FOR BEAMS		
Shuttering(sft)	16488.43	sft
Shuttering rate per sft	55	Rs./sft
FOR COLUMNS		
Shuttering(sft)	4600	sft

Shuttering rate per sft	70	Rs./sft
FOR ROOF		
Shuttering(sft)	1750	sft
Shuttering rate per sft	65	Rs./sft

ACTIVITY DURATION AND LABOR REQUIREMENT

Productivity = Labours hour/unit		
Labor hours =	179.4922667	hrs
8 hours a day		
Duration =	22.43653333	days

ITEM	QUANTITY	RATE	COST	
1-Material				
Cement	1074 bags	Rs.1300/bag	1396200	Rs.
Fine Aggregate	2682.76 cft	Rs.30/cft	80483.05	Rs.
Coarse Aggregate	5365.53 cft	Rs 55/cft	295104.5	Rs.
Water	4975.193 gallons	Rs.1560/1000 gallons	7761.302	Rs.
		Total Material Cost =	**1779549**	**Rs.**
2-Labor Cost				
Skilled	2	Rs.2000/day	89746.13	Rs.
Unskilled	5	Rs.1000/day	112182.7	Rs.
		Total Labor Cost =	**201929**	**Rs.**
3-Equipment Cost				
Shuttering for Beam	42405.048 sft	Rs.55/sft	906863.7	Rs.
Shuttering for Columns	24231.456 sft	Rs.70/sft	322000	Rs.
Shuttering for Roof	7572.33 sft	Rs.65/sft	113750	Rs.
Tools	Lump Sum	Rs.1500/day	33654.8	Rs.
		Total Equipment Cost =	**1376268**	**Rs.**
Total Resource Cost =			**3357746**	**Rs.**
Overheads 5%			167887.3	Rs.
Profit 10%			335774.6	Rs.
Total Price =			**3861408**	**Rs.**
Rate per cft			637.4207	per cft

RATE ANALYSIS FOR 1:3:6 Situ (SUB-STUCTURE)

DATA:

MATERIAL	COST	
Cement	Rs.1300/bag	
Coarse aggregate	Rs.55/cft	
Fine aggregate	Rs.30/cft	
Water Tanker	Rs.1560/1000 gallons	
w/c ratio	0.5	
Labor Productivity	0.8 lhr/cyd	
Skilled Labor	Rs.2000/day	
Unskilled Labor	Rs.1000/day	
Shuttering	Rs.60/sft	
Crew	2 Skilled Labors,4 Unskilled Labors	
Tools	Rs.1500/day	
Overheads	5%	
Profit Margin	10%	

CEMENT BAGS:

Volume =	931.88	cft(Wet Volume)
Volume =	1444.414	cft(Dry Volume)
Cement =	144.4414	cft
Volume of 1 bag =	1.25	cft
No. of bags =	115.55312	
No. of bags =	115.55312	Bags

AGGREGATES:

Fine Aggregates =	433.3242	cft
Coarse Aggregates =	866.6484	cft
Water		
Volume Of Water =	72.2207	cft
1 cft =	7.418	US gallons
Volume Of Water =	535.7331526	Gallons

FORMWORK/SHUTTERING AREA

Shuttering(sft) =	465.94	sft
Shuttering rate per sft	60	Rs/sft

ACTIVITY DURATION AND LABOR REQUIREMENT

Productivity = Labours hour/unit		
Labor hours =	27.61125926	hrs
8 hours a day		
Duration =	3.451407407	days

ITEM	QUANTITY	RATE	COST	
1-Material				
Cement	115.55312	1300	150219.056	Rs.
Fine Aggregate	433.3242	30	12999.726	Rs.
Coarse Aggregate	866.6484	55	47665.662	Rs.
Water	535.7331526	1.56	835.7437181	Rs.
		Total Material	**211720.1877**	**Rs.**
2-Labor Cost				
Skilled	2	Rs.2000/day	13570	Rs.
Unskilled	4	Rs.1000/day	13570	Rs.
		Total Labor Co	**27140**	**Rs.**
3-Equipment Cost				
Shuttering	732.785 sft	Rs.60/sft	43967.1	Rs.
Tools	Lump Sum	1500/day	5088.75	Rs.
		Total Euipmen	**49055.85**	**Rs.**
		Total Resourc	**287916.0377**	**Rs.**
Overheads		5%	14395.80189	Rs.
Profit		10%	28791.60377	Rs.
Total price for 1465.57 cft		**Rs.**	**331103.4434**	**Rs.**
Rate per cft			225.9212753	

RATE ANALYSIS FOR 1:4:8 LEAN CONCRETE(SUPER-STUCTURE)

CEMENT BAGS:

Volume =	1115.29	cft(Wet Volume)
Volume =	1728.6995	cft(Dry Volume)
Cement =	132.9768846	cft
Volume of 1 bag =	1.25	cft
No. of bags =	106.3815077	
No. of bags =	106.3815077	Bags

AGGREGATES:

Fine Aggregates =	531.9075385	cft
Coarse Aggregates =	1063.815077	cft
Water		
Volume Of Water =	66.48844231	cft
1 cft =	7.418	US gallons
Volume Of Water =	493.211265	Gallons

FORMWORK/SHUTTERING AREA

FOR FOOTINGS		
Shuttering(sft)	299.875	sft
Shuttering rate per sft	55	Rs./sft
FOR UGWT		
Shuttering(sft)	51.3225	sft

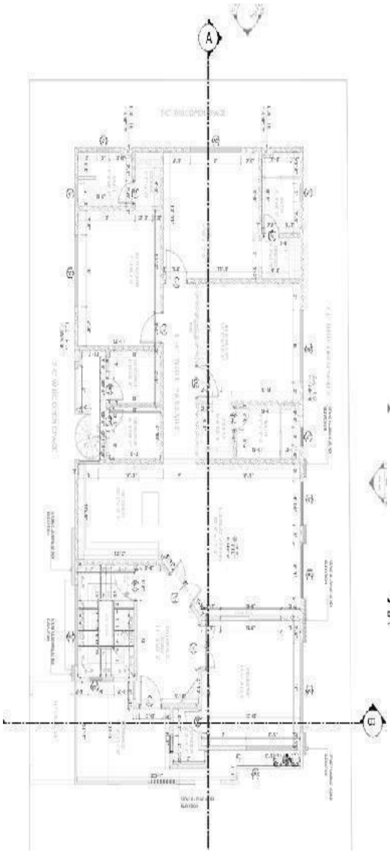
Shuttering rate per sft	70	Rs./sft
FOR WALLS		
Shuttering(sft)	615.9125	sft
Shuttering rate per sft	65	Rs./sft

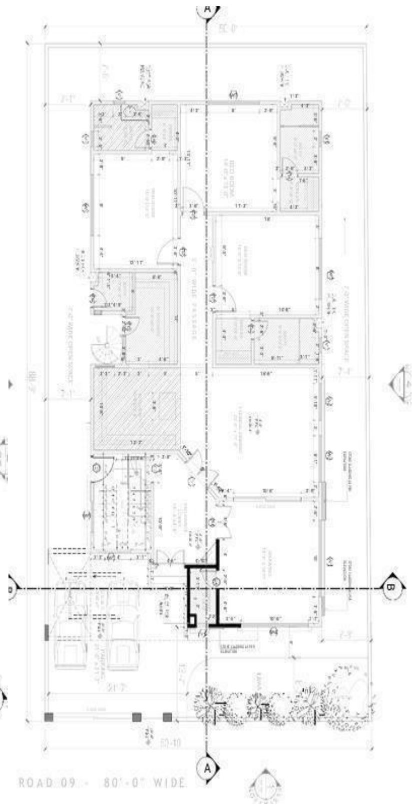
ACTIVITY DURATION AND LABOR REQUIREMENT

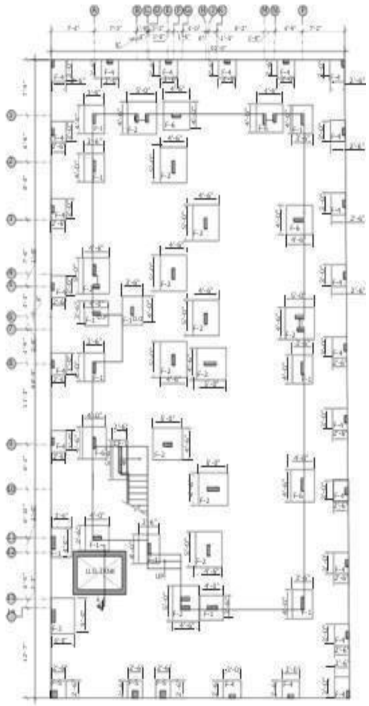
Productivity = Labours hour/unit		
Labor hours =	33.04562963	hrs
8 hours a day		
Duration =	4.130703704	days

ITEM	QUANTITY	RATE	COST	
1-Material				
Cement	106.3815077	1300	138295.96	Rs.
Fine Aggregate	531.9075385	30	15957.22615	Rs.
Coarse Aggregate	1063.815077	55	58509.82923	Rs.
Water	493.211265	1.56	769.4095735	Rs.
		Total Material	**213532.425**	**Rs.**
2-Labor Cost				
Skilled	2	Rs.2000/day	16522.81481	Rs.
Unskilled	5	Rs.1000/day	20653.51852	Rs.
		Total Labor Co	**37176.33333**	**Rs.**
3-Equipment Cost				
Shuttering for Beam	42405.048 sft	Rs.55/sft	16493.125	Rs.
Shuttering for Column	24231.456 sft	Rs.70/sft	3592.575	Rs.
Shuttering for Roof	7572.33 sft	Rs.65/sft	40034.3125	Rs.
Tools	Lump Sum	Rs.1500/day	6196.055556	Rs.
		Total Equipme	**66316.06806**	**Rs.**
Total Resource Cost =			**317024.8263**	**Rs.**
Overheads 5%			15851.24132	Rs.
Profit 10%			31702.48263	Rs.
Total Price =			**364578.5503**	**Rs.**
Rate per cft			326.8912573	per cft

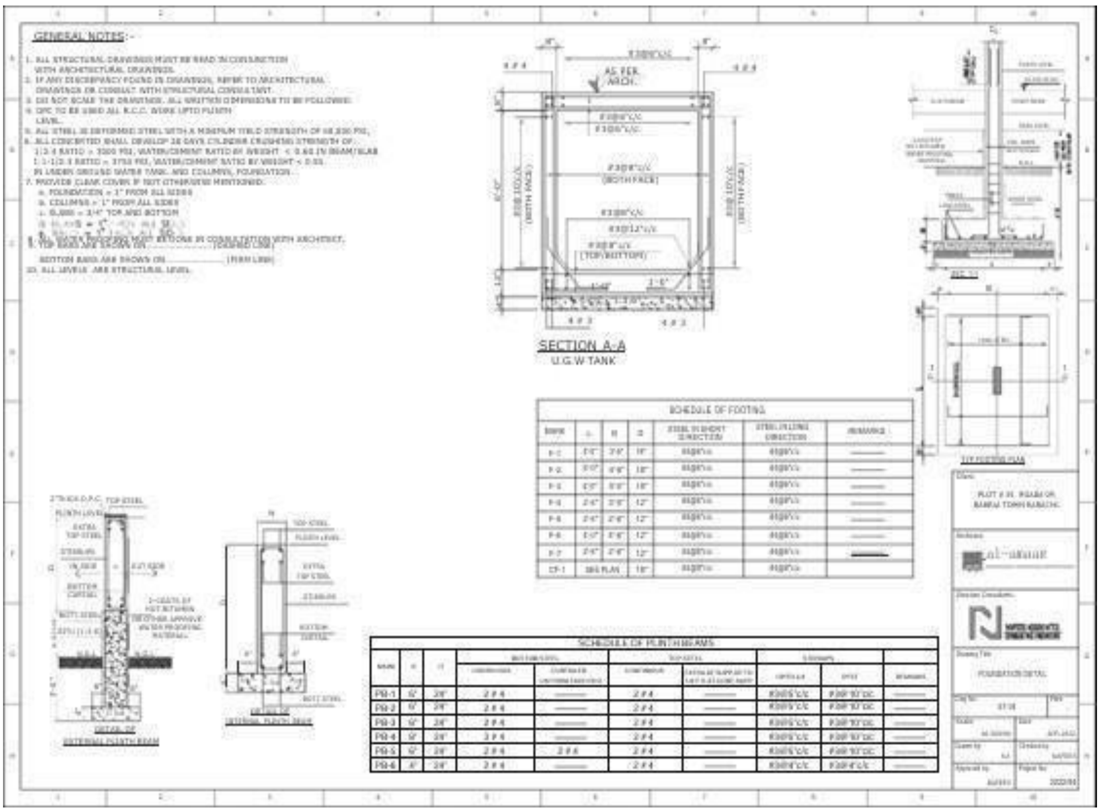
ARCHITECTURAL DRAWING:

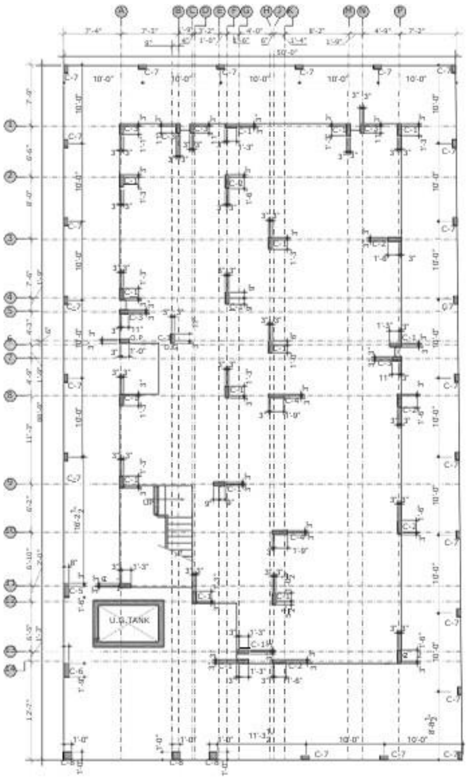






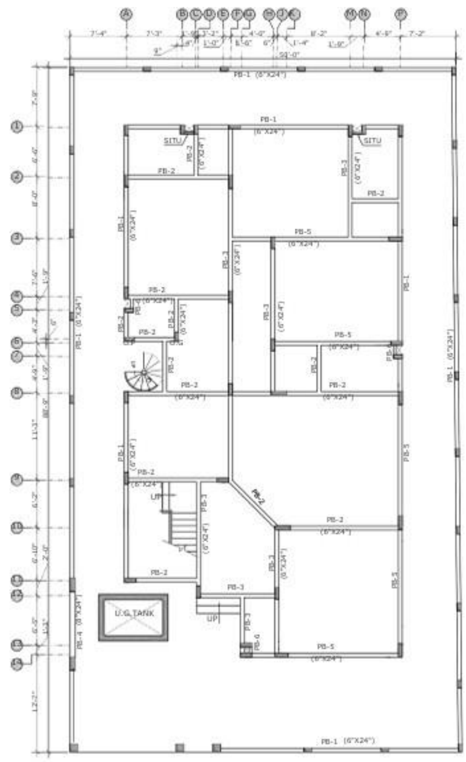
FOUNDATION PLAN



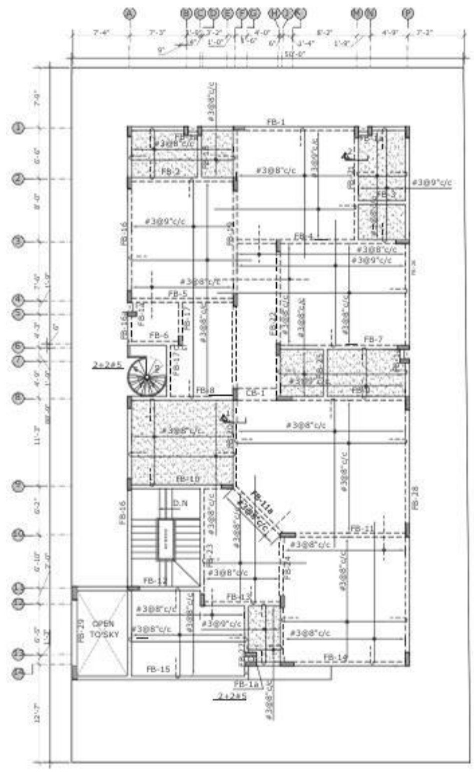


COLUMN LAYOUT PLAN

MARK / FLOOR	C-1	C-2	C-3	C-4	C-5	C-6	C-7	C-8
ROOF TO 1ST FLOOR	6#4 RING #3@8"c/c	6#4 RING #3@8"c/c	4#4 RING #3@8"c/c	8#4 RING #3@8"c/c				
1ST FLOOR TO PLINTH	6#4 RING #3@6"c/c	6#4 RING #3@8"c/c	4#4 RING #3@8"c/c	8#5 RING #3@8"c/c	6#4 RING #3@8"c/c	8#5 RING #3@6"c/c	4#4 UP TO BOUNDARY WALL RING #3@8"c/c	4#4 UP TO BOUNDARY WALL RING #3@8"c/c
PLINTH TO FOUNDATION	6#4 RING #3@8"c/c	6#4 RING #3@8"c/c	4#4 RING #3@8"c/c	8#5 RING #3@8"c/c	6#4 RING #3@8"c/c	8#5 RING #3@6"c/c	4#4 RING #3@8"c/c	4#4 RING #3@8"c/c

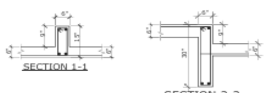


PLINTH BEAM PLAN



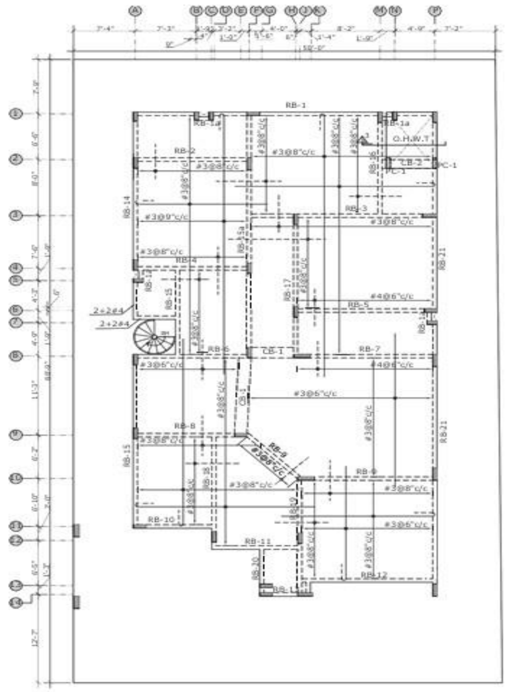
FIRST FLOOR FRAMING PLAN

SCHEDULE OF FIRST FLOOR BEAMS									
MARK	B	H	BOTTOM STEEL		TOP STEEL		STIRRUPS		REMARKS
			CONTINIOUS	CURTAILED L/6 FROM EACH END	CONTINIOUS	EXTRA AT SUPP.UP TO L/4 E.S AT CONT.SUPP	UPTO L/4	REST	
FB-1	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-2	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-3	6"	15"	2 # 4	2 # 4	2 # 4	—	#3@4"c/c	#3@8"c/c	—
FB-4	6"	30"	2 # 4	—	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
FB-5	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-6	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-7	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-8	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-9	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-10	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-11	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-12	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-13	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-14	6"	30"	2 # 4	2 # 4	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
FB-15	6"	24"	2 # 5	2 # 4	2 # 5	2 # 4	#3@4"c/c	#3@8"c/c	—
FB-16	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-17	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-18	6"	15"	2 # 4	—	2 # 4	—	#3@4"c/c	#3@4"c/c	—
FB-19	6"	30"	2 # 4	—	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
FB-20	6"	18"	2 # 4	—	2 # 4	—	#3@4"c/c	#3@8"c/c	—
FB-21	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-22	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-23	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-24	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-25	6"	18"	2 # 4	—	2 # 4	—	#3@4"c/c	#3@4"c/c	—
FB-26	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-27	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-28	6"	30"	2 # 4	2 # 4	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
FB-29	8"	24"	3 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
FB-1a	4"	30"	2 # 4	—	2 # 4	—	#3@4"c/c	#3@4"c/c	—
FB-11a	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
CB-1	18"	6"	4 # 4	—	4 # 4	—	#3@6"c/c	#3@6"c/c	—
FB-16a	6"	30"	2 # 5	—	2 # 5	—	#3@4"c/c	#3@4"c/c	—

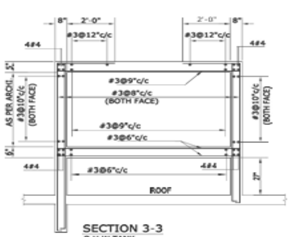

SECTION 1-1

SECTION 2-2

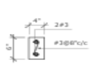
Client:
PLOT # 35, ROAD# 09, BAHRIA TOWN KARACHI.
Architect:
al-amaar
Structure Consultants:
NAFEES ASSOCIATES
Drawing Title:
FIRST FLOOR DETAIL
Dwg No: ST-07
Rev: -
Scale: AS SHOWN
Date: APR 2022
Drawn by: K.A
Checked by: NAFEES
Approved by: NAFEES
Project No: 2022/94


ROOF FRAMING PLAN

SCHEDULE OF ROOF BEAMS									
MARK	B	H	BOTTOM STEEL		TOP STEEL		STIRRUPS		REMARKS
			CONTINIOUS	CURTAILED L/6 FROM EACH END	CONTINIOUS	EXTRA AT SUPP.UP TO L/4 E.S AT CONT.SUPP	UPTO L/4	REST	
RB-1	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-2	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-3	6"	15"	2 # 4	2 # 4	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
RB-4	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-5	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-6	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-7	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-8	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-9	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-10	6"	30"	3 # 4	—	3 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-11	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-12	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-13	6"	30"	2 # 4	—	2 # 4	—	#3@4"c/c	#3@8"c/c	—
RB-14	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-15	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-16	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-17	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-18	6"	30"	2 # 4	2 # 4	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-19	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-20	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-21	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
RB-1a	6"	30"	2 # 4	—	2 # 4	—	#3@5"c/c	#3@10"c/c	—
CB-1	18"	6"	4 # 4	—	4 # 4	—	#3@6"c/c	#3@6"c/c	—
CB-2	15"	6"	3 # 4	—	3 # 4	—	#3@6"c/c	#3@6"c/c	—
RB-15a	6"	30"	2 # 4	—	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—
RB-21a	6"	30"	2 # 4	2 # 4	2 # 4	2 # 4	#3@5"c/c	#3@10"c/c	—

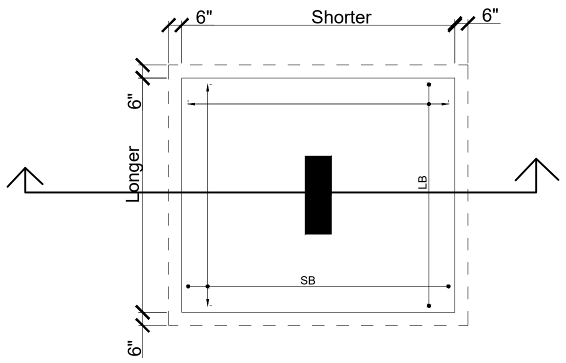


SECTION 3-3
O.H.W.TANK

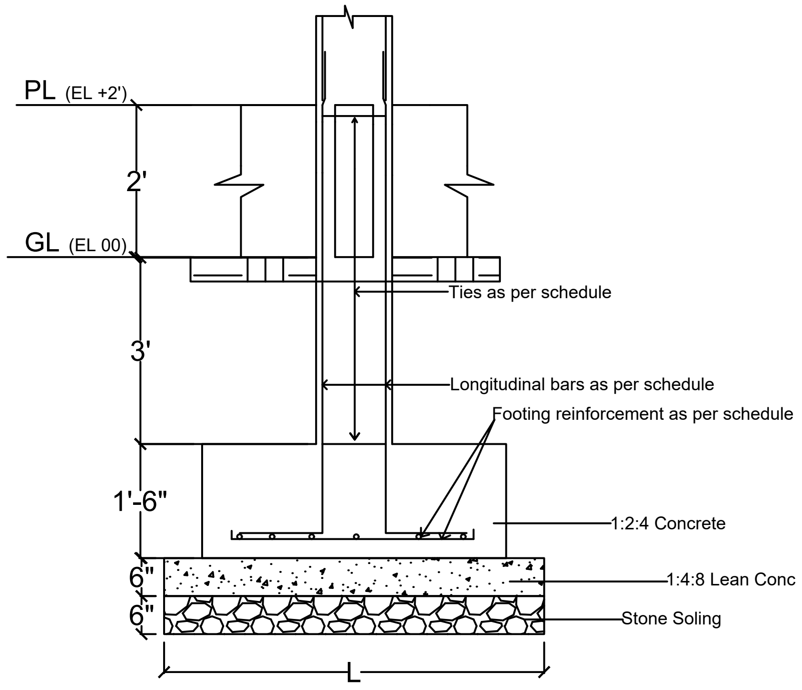

TYP. SECTION
RIB-2

Client:
PLOT # 35, ROAD# 09, BAHRIA TOWN KARACHI.
Architect:
al-amaar
Structure Consultants:
NAFEES ASSOCIATES
Drawing Title:
ROOF DETAIL
Dwg No: ST-08
Rev: -
Scale: AS SHOWN
Date: APR 2022
Drawn by: K.A
Checked by: NAFEES
Approved by: NAFEES
Project No: 2022/94

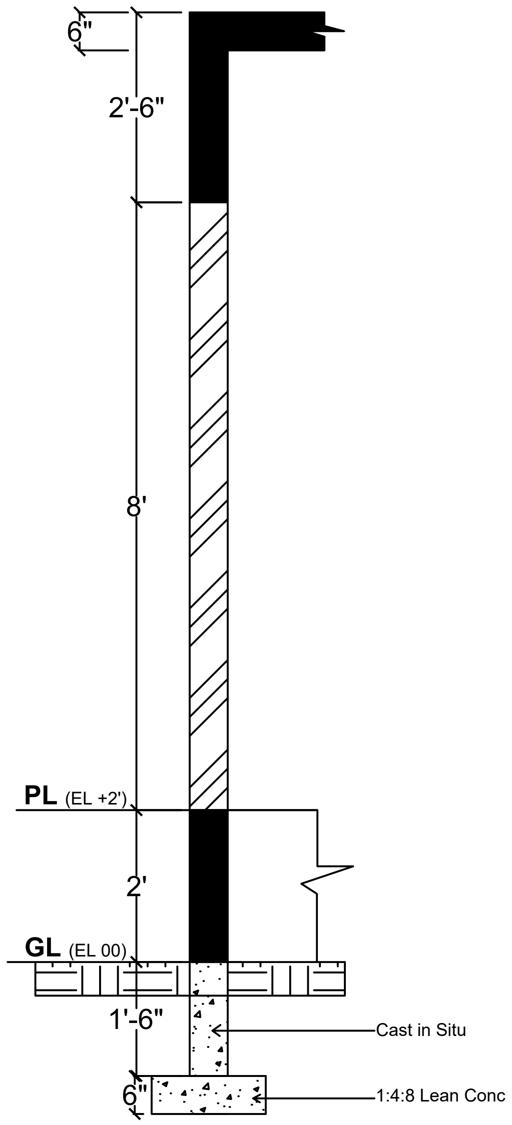
COST CEP SECTIONS



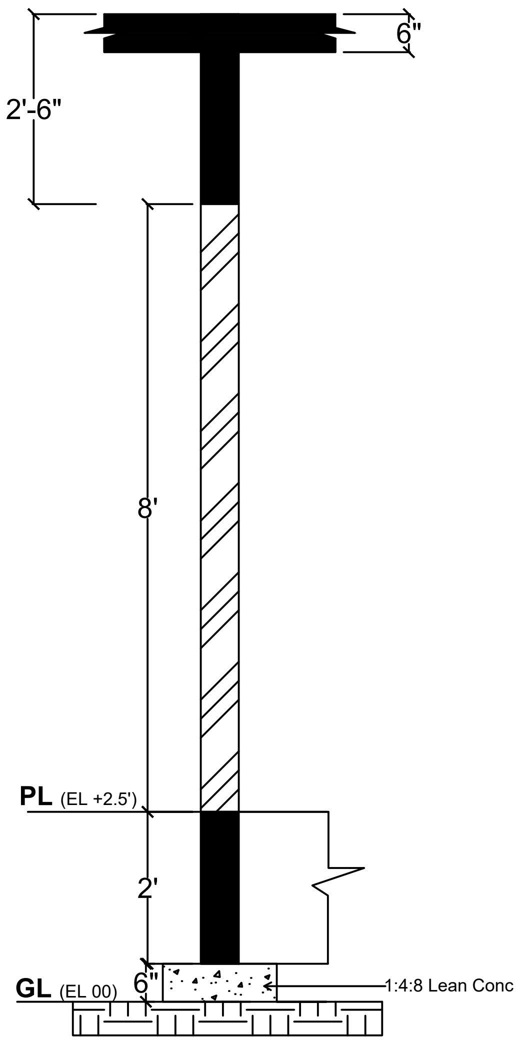
TYP. ISOLATED FOOTING



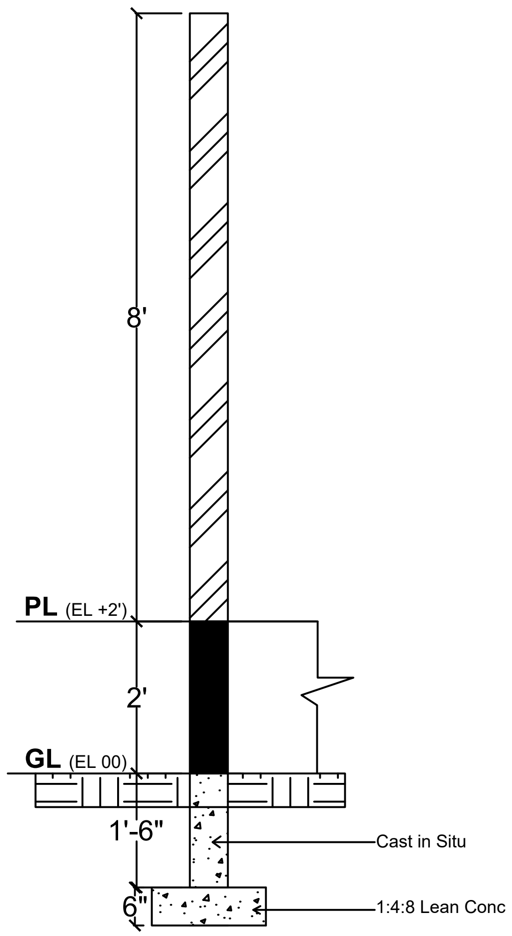
Footing Section



External Wall Section



Internal Wall Section



Boundary Wall Section

General Notes

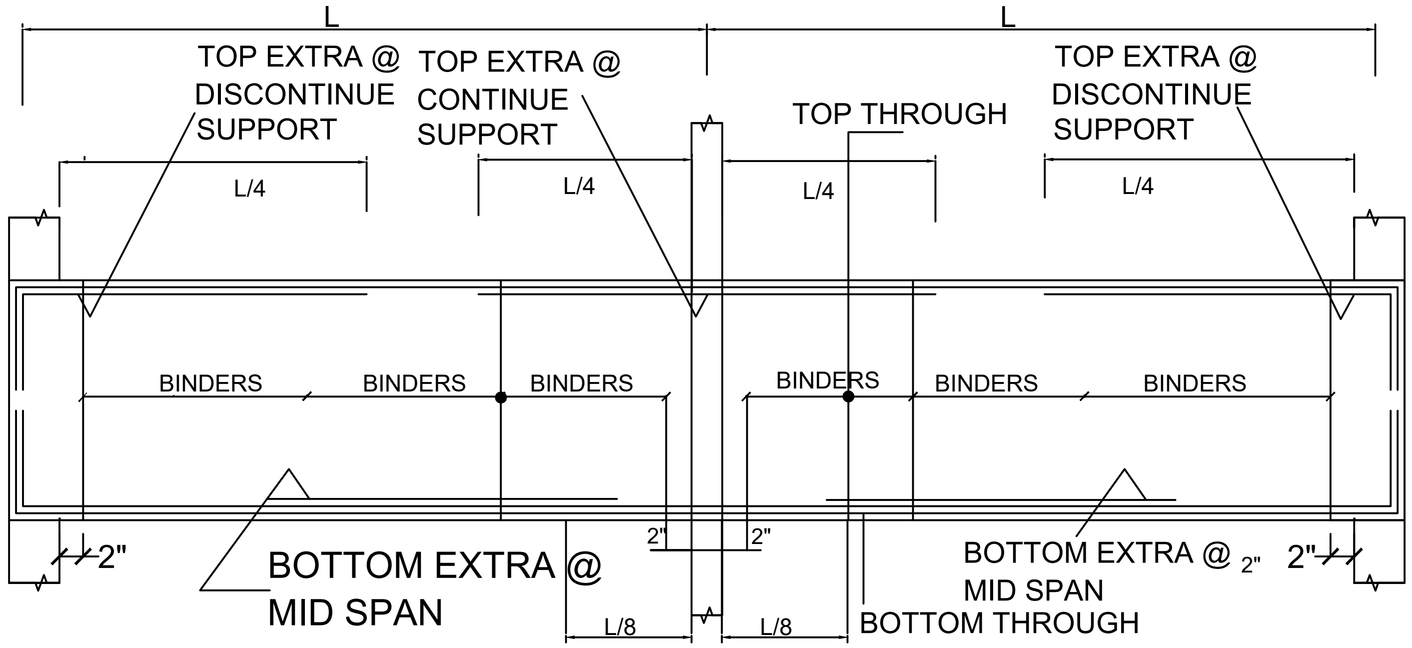
No.	Revision/Issue	Date

Firm Name and Address

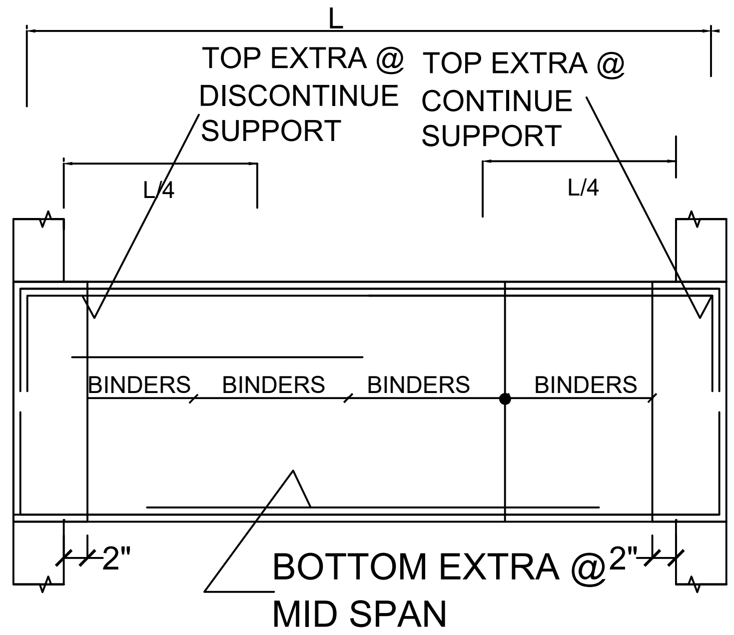
Project Name and Address

Project	Sheet
Date	
Scale	

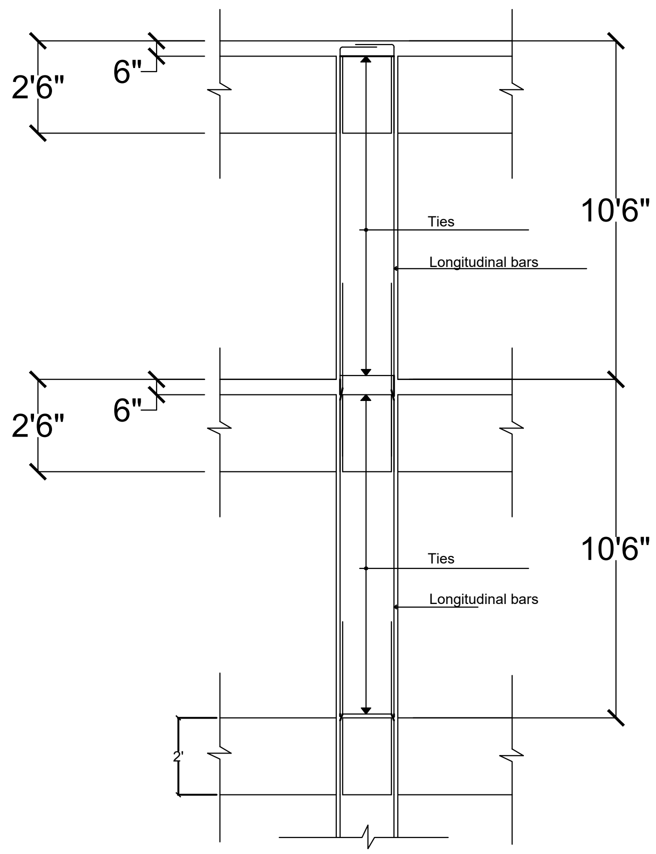
COST CEP SECTIONS



CONTINOUS BEAM ELEVATION



Simply Supported BEAM ELEVATION



COLUMN ELEVATION

General Notes

No.	Revision/Issue	Date

Firm Name and Address

Project Name and Address

Project

Sheet

Date

Scale